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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL SIGNAL PROCESSING (IPCC21EC42)

MODULE-1

DISCRETE FOURIER TRANSFORMS

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MODULE-1 : DISCRETE FOURIER TRANSFORMS (DFT)

To perform frequency analysis on a discrete time sequence $x(n)$, we convert time-domain sequence to an equivalent frequency domain representation.

Applying Fourier Transform on $x(n)$, we get $X(\omega)$, which is continuous and periodic function of frequency. It is not a computationally convenient representation of the sequence $x(n)$.

Representation of a sequence $x(n)$ by samples of its spectrum $X(\omega)$ is known as the Discrete Fourier Transform (DFT).

FREQUENCY DOMAIN SAMPLING :-

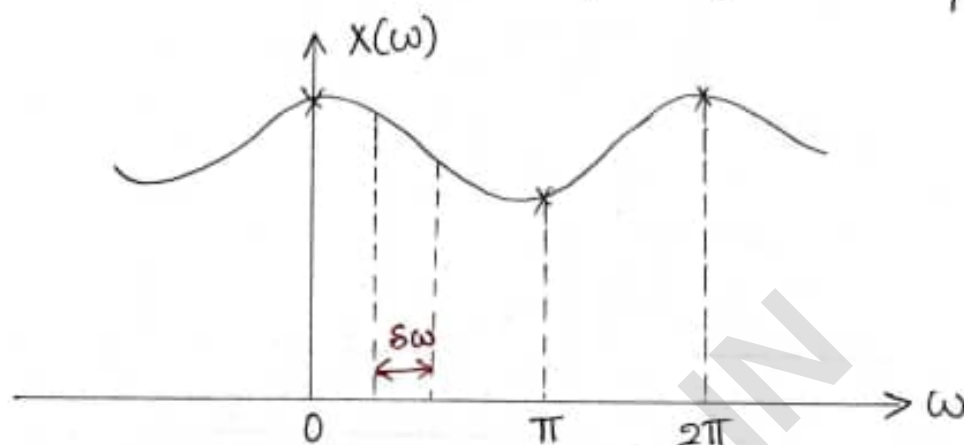
W.K.T aperiodic finite energy signals have continuous spectra.

Fourier Transform (FT) of aperiodic DTS $x(n)$ is given by

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \quad \text{--- (1)}$$

Since $X(\omega)$ is periodic, with period 2π , only samples in the fundamental frequency range are obtained by sampling periodically in frequency at

a spacing of $\delta\omega$ radians between successive samples. We take N equidistant samples in the interval $0 \leq \omega \leq 2\pi$ with spacing $\delta\omega = \frac{2\pi}{N}$.



Substituting $\omega = \frac{2\pi}{N}k$ in eqn (1),

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\frac{2\pi}{N}kn} \quad \text{--- (2)}$$

where $k = 0, 1, 2, \dots, N-1$

Summation in eqn (2) can be subdivided into infinite number of summations, where each sum contains N terms.

$$\therefore X\left(\frac{2\pi}{N}k\right) = \dots + \sum_{n=-N}^{-1} x(n) e^{-j\frac{2\pi}{N}kn} +$$

$$\sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn} + \sum_{n=N}^{2N-1} x(n) e^{-j\frac{2\pi}{N}kn}$$

+ ...

$$= \sum_{l=-\infty}^{+\infty} \sum_{n=lN}^{lN+N-1} x(n) e^{-j\frac{2\pi}{N}kn}$$

If we change the index in the inner summation from n to $n-lN$ and interchange the order of summation, we get,

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=0}^{N-1} \left[\sum_{l=-\infty}^{\infty} x(n-lN) \right] e^{-j\frac{2\pi}{N}k(n-lN)}$$

for $k=0, 1, 2, \dots, N-1$.

$$\text{But } e^{-j\frac{2\pi}{N}k(n-lN)} = e^{-j\frac{2\pi}{N}kn} \cdot e^{+j\frac{2\pi}{N}klN}$$

$$\text{where } e^{j\frac{2\pi}{N}klN} = e^{j2\pi kl} = 1$$

$$\therefore X\left(\frac{2\pi}{N}k\right) = \sum_{n=0}^{N-1} \left[\sum_{l=-\infty}^{+\infty} x(n-lN) \right] e^{-j\frac{2\pi}{N}kn} \quad \text{--- (3)}$$

$$\text{Let } x_p(n) = \sum_{l=-\infty}^{\infty} x(n-lN) \quad \text{--- (4)}$$

$x_p(n)$ is obtained by the periodic repetition of $x(n)$ every N samples. It is periodic with fundamental period N .

$x_p(n)$ can hence be expanded in a Fourier Series as

$$x_p(n) = \sum_{k=0}^{N-1} c_k e^{+j\frac{2\pi}{N}kn} \quad \text{--- (5)}$$

$n=0, 1, 2, \dots, N-1$

with Fourier coefficients,

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p(n) e^{-j\frac{2\pi}{N}kn} \quad \text{--- (6)}$$

$k=0, 1, \dots, N-1$

Comparing eqns (3) & (6),

$$c_k = \frac{1}{N} X\left(\frac{2\pi}{N}k\right) \quad ; k=0, 1, \dots, N-1$$

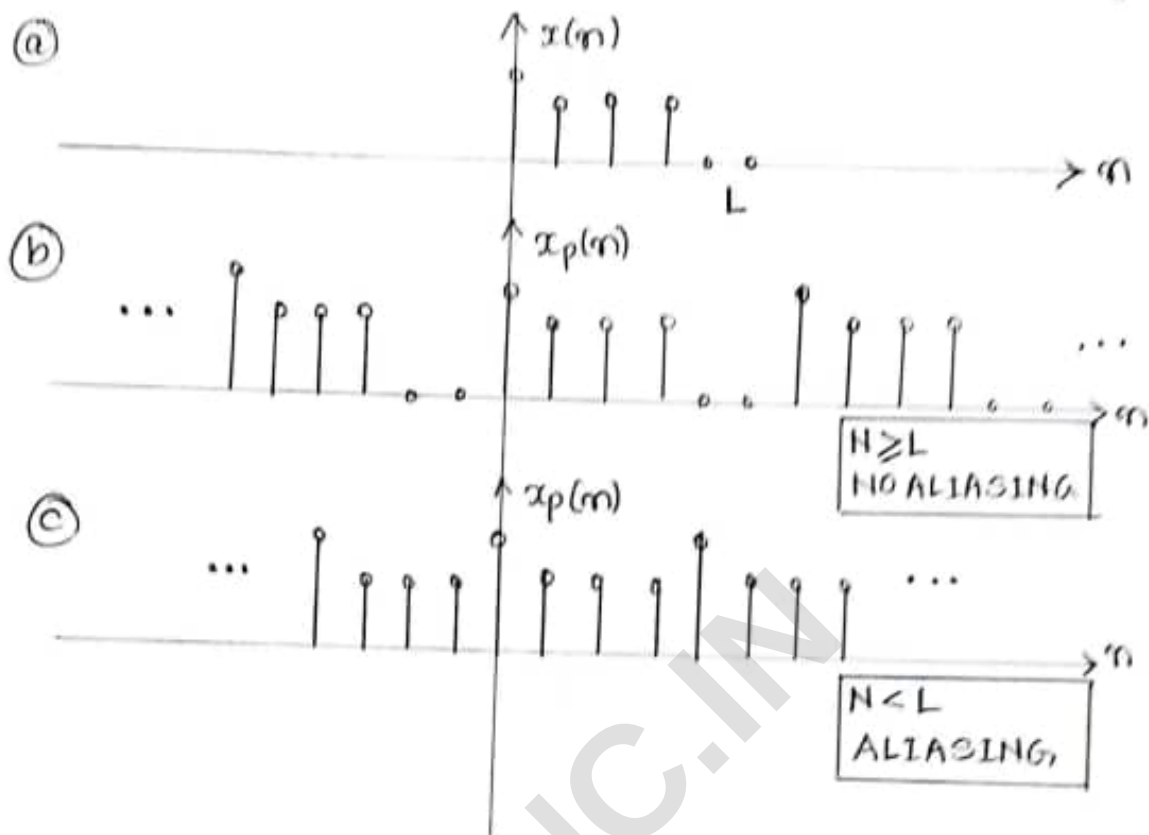
(7)

Substituting eqn (7) in eqn (5),

$$x_p(n) = \frac{1}{N} \sum_{k=0}^{N-1} X\left(\frac{2\pi}{N}k\right) e^{j\frac{2\pi}{N}kn} \quad \text{--- (8)}$$

Equation (8) provides the reconstruction of the periodic signal $x_p(n)$ from the samples of the spectrum $X(\omega)$. However, to recover $X(\omega)$ or $x(n)$ from the samples certain condition needs to be satisfied.

Since $x_p(n)$ is the periodic extension of $x(n)$, it is clear that $x(n)$ can be recovered from $x_p(n)$ if there is no aliasing in the time domain i.e, if $x(n)$ is time limited to less than the period N of $x_p(n)$.



Consider fig (a), $x(n)$ is a sequence non-zero in the interval $0 \leq n \leq L-1$.

When $N \geq L$, $x(n) = x_p(n)$ for $0 \leq n \leq N-1$ [fig (b)]

Hence $x(n)$ can be recovered from $x_p(n)$.

When $N < L$, it cannot be recovered due to time-domain aliasing.

THE DISCRETE FOURIER TRANSFORM (DFT)

In general, the equally spaced frequency samples $X(\frac{2\pi}{N}k)$, $k=0,1,\dots,N-1$, do not uniquely represent the original sequence $x(n)$ when $x(n)$ has infinite duration.

Instead, the frequency samples $X\left(\frac{2\pi k}{N}\right)$ correspond to a periodic sequence $x_p(m)$ of period N , where $x_p(m)$ is an aliased version of $x(m)$ given by

$$x_p(m) = \sum_{l=-\infty}^{\infty} x(m-lN)$$

When the sequence $x(m)$ has a finite duration of length $L \leq N$, then $x_p(m)$ is simply a periodic repetition of $x(m)$, where $x_p(m)$ over a single period is given by

$$x_p(m) = \begin{cases} x(m), & 0 \leq m \leq L-1 \\ 0, & L \leq m \leq N-1 \end{cases}$$

Hence, the frequency samples $X\left(\frac{2\pi k}{N}\right)$, $k=0,1,\dots,N$ uniquely represent the finite duration sequence $x(m)$. Since $x(m) \equiv x_p(m)$ over a single period (padded by $N-L$ zeros), the original finite duration sequence $x(m)$ can be obtained from the frequency samples $X\left(\frac{2\pi k}{N}\right)$ from Eqn (8).

⇒ Summarising: A finite-duration sequence $x(m)$ of length L has a Fourier transform

$$X(\omega) = \sum_{m=0}^{L-1} x(m) e^{-j\omega m}, \quad 0 \leq \omega \leq 2\pi$$

Limits indicate $x(m)$ is zero outside the range $0 \leq m \leq L-1$.

When we sample $X(\omega)$ at equally spaced frequencies $\omega_k = \frac{2\pi k}{N}$, $k=0,1,\dots,N-1$ where $N \geq L$, the samples are

$$X(k) \equiv X\left(\frac{2\pi k}{N}\right) = \sum_{n=0}^{L-1} x(n) e^{-j\frac{2\pi}{N}kn}$$

$$\text{or } X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn}, \quad k=0,1,2,\dots,N-1$$

Upper index in the sum has been increased from $L-1$ to $N-1$ since $x(n)=0$ for $n \geq L$.

The above relation is a formula for transforming a sequence $\{x(n)\}$ of length $L \leq N$ into a sequence of frequency samples $\{X(k)\}$ of length N .

Since, the frequency samples are obtained by evaluating the Fourier transform $X(\omega)$ at a set of N (equally spaced) discrete frequencies, the above relation is called discrete Fourier transform (DFT) of $x(n)$.

Let $W_N = e^{-j\frac{2\pi}{N}}$, then

$$\therefore X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}, \quad k=0,1,2,\dots,N-1$$

The relation which allows us to recover the sequence $x(n)$ from the frequency samples

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N}kn}, \quad n=0,1,2,\dots,N-1$$

$$\text{or } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}$$

is called the Inverse DFT (IDFT).

DFT AS A LINEAR TRANSFORMATION

WKT,

$$\text{DFT} \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$k=0,1,2,\dots,N-1$$

$$\text{IDFT} \{X(k)\} = x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}$$

$$n=0,1,2,\dots,N-1$$

where $W_N = e^{-j\frac{2\pi}{N}}$ Twiddle Factor

From the above equations we observe that the computation of each point of the DFT requires N -complex multiplications and $(N-1)$ complex additions.

Hence, N -point DFT requires N^2 complex multiplications and $N(N-1)$ complex additions.

DFT and IDFT can be viewed as linear transformations on sequences $\{x(n)\}$ and $\{X(k)\}$, respectively.

Let us define a N -point vector x_N of the signal sequence $x(n)$, an N -point vector X_N of frequency samples and a $N \times N$ matrix W_N as

$$x_N = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix} \quad X_N = \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}$$

$$W_N = \begin{bmatrix} W_N^{0.0} & W_N^{0.1} & \dots & W_N^{0(N-1)} \\ W_N^{1.0} & W_N^{1.1} & \dots & W_N^{1(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ W_N^{(N-1)0} & W_N^{(N-1).1} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix}$$

$$\text{or } W_N = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N & W_N^2 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix}$$

From the above definitions, the N -point DFT can be expressed in matrix form as

$$X_N = W_N x_N$$

Where W_N is the matrix of the linear transformation. We observe that W_N is a symmetric matrix. If we assume that inverse of W_N exists then IDFT can be expressed in matrix form as

$$x_N = W_N^{-1} X_N$$

$$\text{or } x_N = \frac{1}{N} W_N^* X_N$$

where W_N^* denotes the complex conjugate of W_N .

PROBLEMS

- 1) Compute the 4-point DFT of a sequence $x(n) = \{1, 2, 3, 4\}$.

Sol'n: — I-METHOD:

From the def'n of DFT wkt,

$$\text{DFT } \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} ; k=0,1,\dots,N-1$$

Given $N=4$,

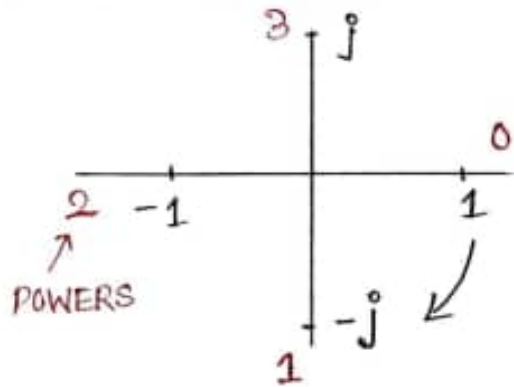
$$\therefore X(k) = \sum_{n=0}^3 x(n) W_4^{kn}$$

$$= x(0) + x(1)W_4^k + x(2)W_4^{2k} + x(3)W_4^{3k}$$

$$X(k) = 1 + 2W_4^k + 3W_4^{2k} + 4W_4^{3k} \quad ; k=0,1,2,3$$

Twiddle Factors:

For $N=4$, First four twiddle factors.



$$W_4^0 = 1 \quad W_4^2 = -1$$

$$W_4^1 = -j \quad W_4^3 = j$$

Alternatively, use $W_N = e^{-j\frac{2\pi}{N}}$ and solve.

$$\text{Let } k=0, \quad X(0) = 1 + 2 + 3 + 4 = \underline{\underline{10}}$$

$$\begin{aligned} \text{Let } k=1, \quad X(1) &= 1 + 2W_4^1 + 3W_4^2 + 4W_4^3 \\ &= 1 + 2(-j) + 3(-1) + 4(j) \\ &= \underline{\underline{-2 + 2j}} \end{aligned}$$

$$\begin{aligned} \text{Let } k=2, \quad X(2) &= 1 + 2W_4^2 + 3W_4^4 + 4W_4^6 \\ &= 1 + 2(-1) + 3(1) + 4(-1) \\ &= \underline{\underline{-2}} \end{aligned}$$

$$\begin{aligned} \text{Let } k=3, \quad X(3) &= 1 + 2W_4^3 + 3W_4^6 + 4W_4^9 \\ &= 1 + 2(j) + 3(-1) + 4(-j) \\ &= \underline{\underline{-2 - 2j}} \end{aligned}$$

$$\therefore X(k) = \{ 10, -2 + 2j, -2, -2 - 2j \}$$

II - METHOD :-

$$X_N = x_N W_N$$

X_N & $x_N \rightarrow$ Row vectors

For $N=4$, $X_4 = x_4 W_4$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4^1 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^4 & W_4^6 \\ 1 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}$$

From periodicity property of W_N : $W_N^{k+N} = W_N^k$

and symmetry property : $W_N^{k+\frac{N}{2}} = -W_N^k$

$$X_4 = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$\therefore X(k) = \{ 10, -2+2j, -2, -2-2j \}$$

Additional problems on DFT :-

2) $x(m) = \{ 0, 1, 2, 3 \}$ for $N=4$.

ANS: $X(k) = \{ 6, -2+2j, -2, -2-2j \}$

3) $x(m) = \{ 1, 2, 2, 1 \}$ for $N=4$.

ANS: $X(k) = \{ 6, -1-j, 0, -1+j \}$

$$4) \quad x(n) = \delta(n) + \delta(n-1) - \delta(n-2) - \delta(n-3) \\ = \{1, 1, -1, -1\}$$

$$\text{ANS : } X(k) = \{0, 2-2j, 0, 2+2j\}$$

5) Find the 4-point IDFT of $X(k) = \{10, -2+2j, -2, -2-2j\}$.

Soln:- From the def'n of IDFT,

$$\text{IDFT} \{X(k)\} = x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \\ n = 0, 1, 2, \dots, N-1$$

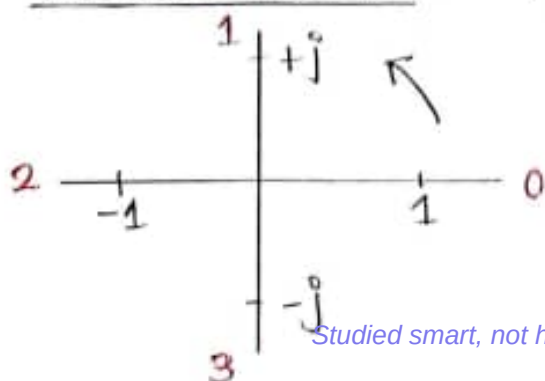
For $N=4$,

$$x(n) = \frac{1}{4} \sum_{k=0}^3 X(k) W_4^{-kn}$$

$$x(n) = \frac{1}{4} \left[X(0) + X(1) W_4^{-n} + X(2) W_4^{-2n} + X(3) W_4^{-3n} \right]$$

$$x(n) = \frac{1}{4} \left[10 + (-2+2j) W_4^{-n} + (-2) W_4^{-2n} + (-2-2j) W_4^{-3n} \right], \quad n = 0, 1, 2, 3.$$

TWIDDLE FACTORS:



For $N=4$,

$$W_4^0 = 1$$

$$W_4^{-1} = j$$

$$W_4^{-2} = -1$$

$$W_4^{-3} = -j$$

Let $n=0$,

$$x(0) = \frac{1}{4} \left[10 - 2 + \cancel{2j} - 2 - 2 - \cancel{2j} \right]$$

$$x(0) = \frac{4}{4} = \underline{\underline{1}}$$

Let $n=1$,

$$x(1) = \frac{1}{4} \left[10 + (-2+2j) W_4^{-1} + (-2) W_4^{-2} + (-2-2j) W_4^{-3} \right]$$

$$= \frac{1}{4} \left[10 + (-2+2j)(j) + (-2)(-1) + (-2-2j)(-j) \right]$$

$$x(1) = \frac{8}{4} = \underline{\underline{2}}$$

Let $n=2$,

$$x(2) = \frac{1}{4} \left[10 + (-2+2j) W_4^{-2} + (-2) W_4^{-4} + (-2-2j) W_4^{-6} \right]$$

$$= \frac{1}{4} \left[10 + (-2+2j)(-1) + (-2)(1) + (-2-2j)(-1) \right]$$

$$x(2) = \frac{12}{4} = \underline{\underline{3}}$$

Let $n=3$,
$$x(3) = \frac{1}{4} \left[10 + (-2+2j) W_4^{-3} + (-2) W_4^{-6} + (-2-2j) W_4^{-9} \right]$$

$$x(3) = \frac{16}{4} = \underline{\underline{4}}$$

$$\therefore x(n) = \{ 1, 2, 3, 4 \}$$

II - METHOD :-

$$\text{WKT, } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}$$

Above equation can also be written as,

$$X_N = \frac{1}{N} X_N W_N^* \quad X_N, X_N \rightarrow \text{Row vectors.}$$

Given $N=4$,

$$X_4 = \frac{1}{4} X_4 W_4^*$$

$$X_4 = \frac{1}{4} \begin{bmatrix} 10 & (-2+2j) & -2 & (-2-2j) \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 10-2+2j-2-2-2j \\ 10-2j-2+2+2j-2 \\ 10+2-2j-2+2+2j \\ 10+2j+2+2-2j+2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 \\ 8 \\ 12 \\ 16 \end{bmatrix}$$

$$\therefore x(n) = \{1, 2, 3, 4\}$$

NOTE : ANSWER
IS IN ROW,
REPRESENTED AS
COLUMN ONLY FOR
CONVENIENCE.

⑥ Find the DFT of $x(n) = \delta(n)$.

Soln:- From def'n of DFT,

$$\text{DFT} \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$\text{Hence, DFT} \{ \delta(n) \} = \sum_{n=0}^{N-1} \delta(n) W_N^{kn}$$

$$\text{W.K.T, } \delta(n) = \begin{cases} 1, & \text{at } n=0 \\ 0, & \text{for } n \neq 0 \end{cases}$$

$$\therefore \text{DFT} \{ \delta(n) \} = \delta(0) W_N^0 = 1 //$$

$$\text{or } \boxed{\delta(n) \xleftrightarrow{\text{N-PT DFT}} 1}$$

⑦ Find the DFT of $x(n) = \delta(n - n_0)$.

Soln:- From the def'n of DFT,

$$\text{DFT} \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$\text{Hence, DFT} \{ \delta(n - n_0) \} = \sum_{n=0}^{N-1} \delta(n - n_0) W_N^{kn}$$

$$\text{W.K.T, } \delta(n - n_0) = \begin{cases} 1, & \text{at } n = n_0 \\ 0, & \text{for } n \neq n_0 \end{cases}$$

$$\therefore \text{DFT} \{ \delta(n - n_0) \} = \delta(0) W_N^{kn_0} = W_N^{kn_0}$$

$$\boxed{\delta(n - n_0) \xleftrightarrow{\text{DFT}} W_N^{kn_0}}$$

⑧ Find N-point DFT of $x(n) = e^{j\frac{2\pi}{N}k_0n}$.

Soln:- From definition of DFT

$$\text{DFT } \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$\text{Hence, DFT } \left\{ e^{j\frac{2\pi}{N}k_0n} \right\} = \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}k_0n} \cdot e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} \left[e^{-j\frac{2\pi}{N}(k-k_0)n} \right]$$

$$X(k) = \frac{1 - \left[e^{-j\frac{2\pi}{N}(k-k_0)N} \right]}{1 - e^{-j\frac{2\pi}{N}(k-k_0)}} \quad \because \sum_{n=0}^N a^n = \frac{1 - a^{N+1}}{1 - a}$$

at $k=k_0$, $X(k) = \frac{0}{0}$, Using L-Hospital's Method,

$$= \frac{0 - \left(-j\frac{2\pi}{N} e^{-j\frac{2\pi}{N}(k-k_0)N} \right)}{0 - \left(-j\frac{2\pi}{N} e^{-j\frac{2\pi}{N}(k-k_0)N} \right)} \bigg|_{k=k_0}$$

At $k=k_0$,

$$= \frac{j\frac{2\pi}{N} \cdot e^0}{j\frac{2\pi}{N} \cdot e^0} = N$$

$\therefore X(k) = N$ at $k=k_0$

at $k \neq k_0$ $X(k) = 0$

$$\therefore \boxed{X(k) = N \delta(k-k_0)} \text{ or } \boxed{X(k) = \begin{cases} N & \text{at } k=k_0 \\ 0 & \text{for } k \neq k_0 \end{cases}}$$

⑨ Find the DFT of $x(m) = e^{-j\frac{2\pi}{N}k_0 m}$.

Soln:- From the def'n of DFT,

$$\text{DFT } \{x(m)\} = X(k) = \sum_{m=0}^{N-1} x(m) W_N^{km}$$

$$\text{Hence, DFT } \left\{ e^{-j\frac{2\pi}{N}k_0 m} \right\} = \sum_{m=0}^{N-1} e^{-j\frac{2\pi}{N}k_0 m} \cdot e^{-j\frac{2\pi}{N}km}$$

$$= \sum_{m=0}^{N-1} \left[e^{-j\frac{2\pi}{N}(k+k_0)} \right]^m$$

$$= \frac{1 - \left[e^{-j\frac{2\pi}{N}(k+k_0)} \right]^N}{1 - e^{-j\frac{2\pi}{N}(k+k_0)}} \quad \because \sum_{m=0}^N a^m = \frac{1 - a^{N+1}}{1 - a}$$

$$\text{at } k = -k_0, X(k) = \frac{0}{0}.$$

Using L-Hospital's Rule,

$$X(k) = \frac{0 - (-j2\pi) e^{-j2\pi(k+k_0)}}{0 - (-j\frac{2\pi}{N}) e^{-j\frac{2\pi}{N}(k+k_0)}} \bigg|_{k=-k_0}$$

$$\text{at } k = -k_0, X(k) = \frac{j2\pi}{j2\pi/N} = N$$

$$\text{at } k \neq -k_0, X(k) = 0$$

$$\therefore \boxed{X(k) = N \delta(k+k_0)}$$

$$\text{or } \boxed{X(k) = \begin{cases} N & \text{at } k = -k_0 \\ 0 & \text{at } k \neq -k_0 \end{cases}}$$

$$\text{or } X(k) = N \delta(k - N + k_0)$$

⑩ Find the N-point DFT of $x(m) = \cos \frac{2\pi}{N} k_0 m$

Sol'n:- From def'n of DFT,

$$\text{DFT } \{x(m)\} = X(k) = \sum_{n=0}^{N-1} x(m) W_N^{km}$$

$$\text{DFT } \left\{ \cos\left(\frac{2\pi}{N} k_0 m\right) \right\} = \sum_{n=0}^{N-1} \cos\left(\frac{2\pi}{N} k_0 m\right) W_N^{km}$$

$$= \sum_{n=0}^{N-1} \left[\frac{e^{j\frac{2\pi}{N} k_0 m} + e^{-j\frac{2\pi}{N} k_0 m}}{2} \right] e^{-j\frac{2\pi}{N} km}$$

$$= \frac{1}{2} \left\{ \sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N} (k-k_0) m} + \sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N} (k+k_0) m} \right\}$$

$$\because \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$= \frac{1}{2} \{ N \delta(k-k_0) + N \delta(k+k_0) \}$$

$$\therefore \boxed{\cos\left(\frac{2\pi}{N} k_0 m\right) \xleftrightarrow{\text{DFT}} \frac{N}{2} [\delta(k-k_0) + \delta(k+k_0)]}$$

⑪ Similar problem, $x(m) = \sin \frac{2\pi}{N} k_0 m$

ANS: $\sin\left(\frac{2\pi}{N} k_0 m\right) \xleftrightarrow{\text{DFT}} \frac{N}{2j} [\delta(k-k_0) - \delta(k+k_0)]$

(12) Find the 5-point DFT of $x(n) = \{1, 1, 1\}$.

Sol'n: From the def'n of DFT, WKT,

$$\text{DFT } \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$k = 0, 1, \dots, N-1$

Given $N = 5$,

$$X(k) = \sum_{n=0}^4 x(n) W_5^{kn} ; k = 0, 1, 2, 3, 4$$

$$X(k) = x(0) + x(1) W_5^k + x(2) W_5^{2k} + x(3) W_5^{3k} + x(4) W_5^{4k}$$

$$X(k) = 1 + W_5^k + W_5^{2k} \quad k = 0, 1, 2, 3, 4$$

Twiddle Factors: $N = 5$

$$W_5^0 = 1$$

$$W_5^1 = e^{-j\frac{2\pi}{5}} = \cos\left(\frac{2\pi}{5}\right) - j \sin\left(\frac{2\pi}{5}\right)$$

$$= 0.3090 - j 0.95$$

$$W_5^2 = e^{-j\frac{2\pi}{5} \cdot 2} = \cos\left(\frac{4\pi}{5}\right) - j \sin\left(\frac{4\pi}{5}\right)$$

$$= -0.809 - j 0.5878$$

$$W_5^3 = e^{-j\frac{2\pi}{5} \cdot 3} = \cos\left(\frac{6\pi}{5}\right) - j \sin\left(\frac{6\pi}{5}\right)$$

$$= -0.809 + j 0.587$$

$$W_5^4 = e^{-j\frac{2\pi}{5} \cdot 4} = \cos\left(\frac{8\pi}{5}\right) - j \sin\left(\frac{8\pi}{5}\right) \\ = 0.309 + j 0.95$$

Evaluate $X(k)$,

$$\text{at } k=0, \quad X(0) = 1 + 1 + 1 = \underline{\underline{3}}$$

$$\text{at } k=1, \quad X(1) = 1 + W_5^1 + W_5^2 = \underline{\underline{0.5 - j 1.5388}}$$

$$\text{at } k=2, \quad X(2) = 1 + W_5^2 + W_5^4 = \underline{\underline{0.5 + j 0.3633}}$$

$$\text{at } k=3, \quad X(3) = 1 + W_5^3 + W_5^6 = \underline{\underline{0.5 - j 0.3633}}$$

$$\text{at } k=4, \quad X(4) = 1 + W_5^4 + W_5^8 = \underline{\underline{0.5 + j 1.5388}}$$

$$X(k) = \left\{ 3, 0.5 - j 1.5388, 0.5 + j 0.3633, \right. \\ \left. 0.5 - j 0.3633, 0.5 + j 1.5388 \right\}$$

⑬ Find IDFT for the following sequence

$$X(k) = \{ 5, 0, (1-j), 0, 1, 0, (1+j), 0 \}$$

Sol'n:- Length of the sequence is 8.

$$\therefore N = 8$$

From the def'n of IDFT,

$$\text{IDFT } \{ X(k) \} = x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-km}$$

where $m = 0, 1, 2, \dots, N-1$.

For $N=8$,

$$x(m) = \frac{1}{8} \sum_{k=0}^7 X(k) W_8^{-km}$$

$$x(m) = \frac{1}{8} \left[X(0) W_8^0 + X(1) W_8^{-m} + X(2) W_8^{-2m} + X(3) W_8^{-3m} + X(4) W_8^{-4m} + X(5) W_8^{-5m} + X(6) W_8^{-6m} + X(7) W_8^{-7m} \right]$$

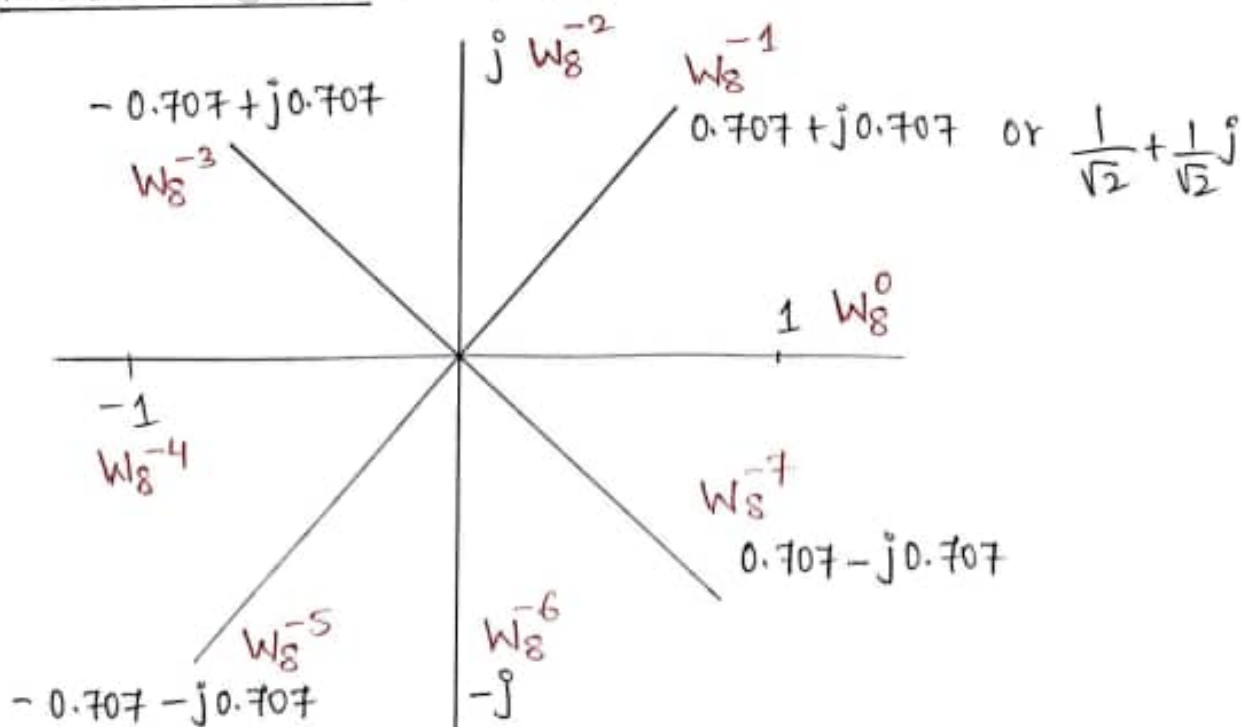
$$x(m) = \frac{1}{8} \left[5 + 0 + (1-j) W_8^{-2m} + 0 + W_8^{-4m} + 0 + (1+j) W_8^{-6m} + 0 \right]$$

where $m = 0, 1, 2, \dots, 7$

$$\text{or } x(m) = \frac{1}{8} \left[5 + (1-j) W_8^{-2m} + W_8^{-4m} + (1+j) W_8^{-6m} \right]$$

TWIDDLE FACTORS :

$N=8$



$$\begin{aligned}\text{For } n=0, \quad x(0) &= \frac{1}{8} \{ 5 + 1-j + 1 + 1+j \} \\ &= \frac{8}{8} = \underline{\underline{1}}\end{aligned}$$

$$\begin{aligned}\text{For } n=1, \quad x(1) &= \frac{1}{8} \{ 5 + (1-j)W_8^{-2} + W_8^{-4} \\ &\quad + (1+j)W_8^{-6} \} \\ &= \frac{1}{8} \{ 5 + (1-j)j + (-1) + (1+j)(-j) \} \\ &= \frac{1}{8} \{ 5 + j - j^2 - 1 - j - j^2 \} \\ &= \frac{6}{8} = \underline{\underline{\frac{3}{4}}}\end{aligned}$$

$$\begin{aligned}\text{For } n=2, \quad x(2) &= \frac{1}{8} \{ 5 + (1-j)W_8^{-4} + W_8^{-8} \\ &\quad + (1+j)W_8^{-12} \} \\ &= \frac{1}{8} \{ 5 + (1-j)(-1) + 1 + (1+j)(-1) \} \\ &= \frac{1}{8} \{ 5 - 1 + j + 1 - 1 - j \} \\ &= \frac{4}{8} = \underline{\underline{\frac{1}{2}}}\end{aligned}$$

$$\begin{aligned}\text{For } n=3, \quad x(3) &= \frac{1}{8} \{ 5 + (1-j)W_8^{-6} + W_8^{-12} + \\ &\quad (1+j)W_8^{-18} \} \\ &= \frac{1}{8} \{ 5 + (1-j)(-j) + (-1) + (1+j)(j) \}\end{aligned}$$

$$\begin{aligned}
 x(3) &= \frac{1}{8} \{ 5 - j + j^2 - 1 + j + j^2 \} \\
 &= \frac{1}{8} \{ 5 - j - 1 - 1 + j - 1 \} = \frac{2}{8} = \underline{\underline{\frac{1}{4}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{for } n=4, \quad x(4) &= \frac{1}{8} \{ 5 + (1-j) W_8^{-8} + W_8^{-16} + (1+j) W_8^{-24} \} \\
 &= \frac{1}{8} \{ 5 + (1-j) + 1 + (1+j) \} \\
 &= \frac{8}{8} = \underline{\underline{1}}
 \end{aligned}$$

$$\begin{aligned}
 \text{for } n=5, \quad x(5) &= \frac{1}{8} \{ 5 + (1-j) W_8^{-10} + W_8^{-20} + (1+j) W_8^{-30} \} \\
 &= \frac{1}{8} \{ 5 + (1-j)j + (-1) + (1+j)(-j) \} \\
 &= \frac{1}{8} \{ 5 + j - j^2 - 1 - j - j^2 \} \\
 &= \frac{6}{8} = \underline{\underline{\frac{3}{4}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{for } n=6, \quad x(6) &= \frac{1}{8} \{ 5 + (1-j) W_8^{-12} + W_8^{-24} + (1+j) W_8^{-36} \} \\
 &= \frac{1}{8} \{ 5 + (1-j)(-1) + 1 + (1+j)(-1) \} \\
 &= \frac{1}{8} \{ 5 - 1 + j + 1 - 1 - j \} = \frac{4}{8} = \underline{\underline{\frac{1}{2}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{for } m=7, \quad x(7) &= \frac{1}{8} \left\{ 5 + (1-j) W_8^{-14} + W_8^{-28} \right. \\
 &\quad \left. + (1+j) W_8^{-42} \right\} \\
 &= \frac{1}{8} \left\{ 5 + (1-j)(-j) + (-1) + (1+j)(j) \right\} \\
 &= \frac{1}{8} \left\{ 5 - j + j^2 - 1 + j + j^2 \right\} \\
 &= \frac{2}{8} = \underline{\underline{\frac{1}{4}}}
 \end{aligned}$$

$$\therefore x(m) = \left\{ 1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, 1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4} \right\}$$

- (14) Find DFT of the sequence $x(m) = \begin{cases} 1, & 0 \leq m \leq 2 \\ 0, & \text{otherwise} \end{cases}$
for $N=8$. Plot $|X(k)|$ & $\angle X(k)$.

Sol'n:- From the def'n of DFT,

$$\text{DFT } \{x(m)\} = X(k) = \sum_{m=0}^{N-1} x(m) W_N^{km}$$

for $N=8$,

$$X(k) = \sum_{m=0}^7 x(m) W_8^{km}, \quad 0 \leq k \leq 7$$

$$\text{Given } x(m) = \{1, 1, 1\}$$

$$\therefore X(k) = 1 + W_8^k + W_8^{2k} + 0 + 0 + 0 + 0 + 0$$

$$\boxed{X(k) = 1 + W_8^k + W_8^{2k}}, k = 0, 1, 2, \dots, 7$$

Twiddle Factors for $N=8$

$$W_8^0 = 1$$

$$W_8^4 = -1$$

$$W_8^1 = 0.707 - j0.707$$

$$W_8^5 = -0.707 + j0.707$$

$$W_8^2 = -j$$

$$W_8^6 = j$$

$$W_8^3 = -0.707 - j0.707$$

$$W_8^7 = 0.707 + j0.707$$

(Locate points on the unit circle in complex plane direction is clockwise for DFT)

$$\text{for } k=0, \quad X(0) = 1 + 1 + 1 = \underline{\underline{3}}$$

$$\begin{aligned} \text{for } k=1, \quad X(1) &= 1 + W_8^1 + W_8^2 \\ &= 1 + (0.707 - j0.707) + (-j) \\ &= \underline{\underline{1.707 - j1.707}} \end{aligned}$$

$$\begin{aligned} \text{for } k=2, \quad X(2) &= 1 + W_8^2 + W_8^4 \\ &= 1 + (-j) + (-1) = \underline{\underline{-j}} \end{aligned}$$

$$\begin{aligned} \text{for } k=3, \quad X(3) &= 1 + W_8^3 + W_8^6 \\ &= 1 + (-0.707 - j0.707) + j \\ &= \underline{\underline{0.293 + j0.293}} \end{aligned}$$

$$\begin{aligned} \text{for } k=4, \quad X(4) &= 1 + W_8^4 + W_8^8 \\ &= 1 + (-1) + 1 = \underline{\underline{1}} \end{aligned}$$

$$\begin{aligned} \text{for } k=5, \quad X(5) &= 1 + W_8^5 + W_8^{10} \\ &= 1 + (-0.707 + j 0.707) + (-j) \\ &= \underline{\underline{0.293 - j 0.293}} \end{aligned}$$

$$\begin{aligned} \text{for } k=6, \quad X(6) &= 1 + W_8^6 + W_8^{12} \\ &= 1 + j + (-1) = \underline{\underline{j}} \end{aligned}$$

$$\begin{aligned} \text{for } k=7, \quad X(7) &= 1 + W_8^7 + W_8^{14} \\ &= 1 + 0.707 + j 0.707 + j \\ &= \underline{\underline{1.707 + j 1.707}} \end{aligned}$$

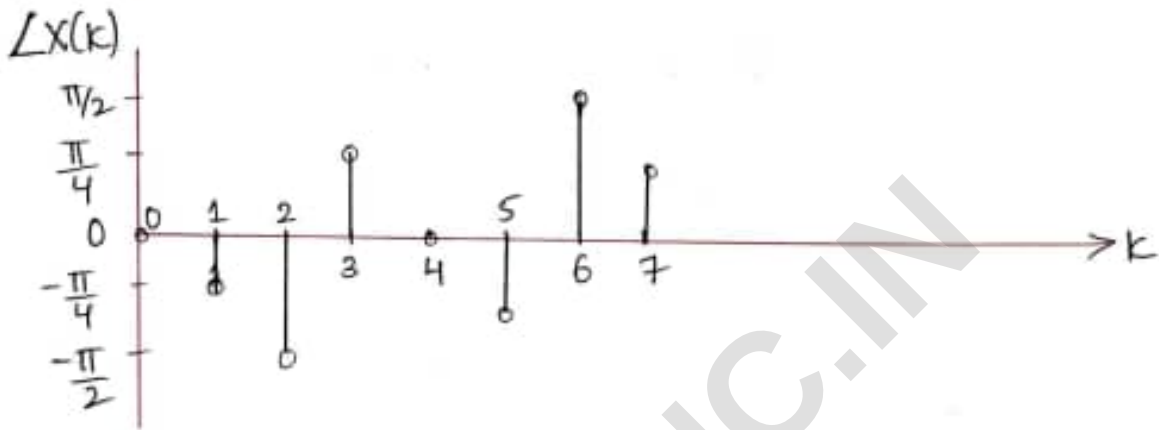
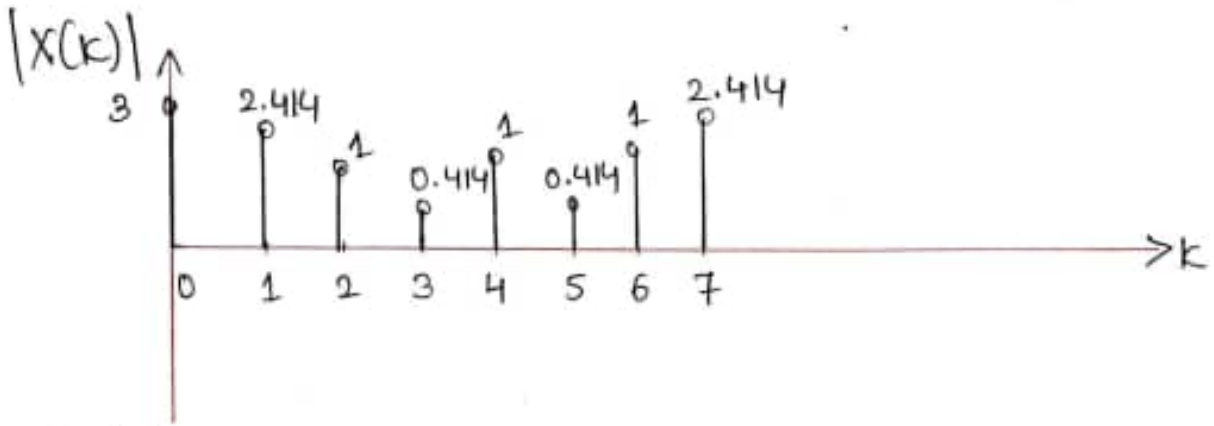
$$\therefore X(k) = \left\{ 3, (1.707 - j 1.707), -j, (0.293 + j 0.293), 1, (0.293 - j 0.293), j, (1.707 + j 1.707) \right\}$$

For magnitude Plot,

$$|X(k)| = \left\{ 3, 2.414, 1, 0.414, 1, 0.414, 1, 2.414 \right\}$$

Phase Plot,

$$\angle X(k) = \left\{ 0, -\frac{\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4} \right\}$$



PROPERTIES OF DFT

1) LINEARITY :-

STATEMENT:

$$\text{If } x_1(n) \xleftrightarrow{\text{DFT}} X_1(k)$$

$$x_2(n) \xleftrightarrow{\text{DFT}} X_2(k)$$

$$\text{then } ax_1(n) + bx_2(n) \xleftrightarrow{\text{DFT}} aX_1(k) + bX_2(k)$$

Proof: W.K.T., $X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

Let $x(n) = x_1(n)$,

$$\text{then } X_1(k) = \sum_{n=0}^{N-1} x_1(n) W_N^{kn} \quad \text{--- (1)}$$

Let $x(n) = x_2(n)$,

$$\text{then } X_2(k) = \sum_{n=0}^{N-1} x_2(n) W_N^{kn} \quad \text{--- (2)}$$

Let $x(n) = ax_1(n) + bx_2(n)$, then

$$\text{DFT } \{ ax_1(n) + bx_2(n) \} = \sum_{n=0}^{N-1} [ax_1(n) + bx_2(n)] W_N^{kn}$$

$$= a \sum_{n=0}^{N-1} x_1(n) W_N^{kn} + b \sum_{n=0}^{N-1} x_2(n) W_N^{kn}$$

$$= aX_1(k) + bX_2(k)$$

Hence the proof.

2) PERIODICITY

STATEMENT: If $x(n) \xleftrightarrow{\text{DFT}} X(k)$

$$\text{then } x(n+N) = x(n)$$

$$X(k+N) = X(k)$$

$$\text{or } x(n+N) \xleftrightarrow{\text{DFT}} X(k)$$

Proof: a) TPT $x(n+N) = x(n)$

$$\text{W.K.T } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}$$

replace n by $n+N$,

$$x(n+N) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-k(n+N)}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \cdot W_N^{-kN}$$

$$\text{W.K.T } W_N^{-kN} = 1$$

$$\therefore x(n+N) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}$$

$$\boxed{x(n+N) = x(n)}$$

Hence proved.

b) T.P.T $X(k+N) = X(k)$

WKT $X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

replace k by $k+N$,

$$X(k+N) = \sum_{n=0}^{N-1} x(n) W_N^{(k+N)n}$$

$$= \sum_{n=0}^{N-1} x(n) W_N^{kn} \cdot W_N^{Nn}$$

$$X(k+N) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad \because W_N^{Nn} = 1$$

$$\therefore \boxed{X(k+N) = X(k)}$$

Hence proved.

c) T.P.T $x(n+N) \xleftrightarrow{\text{DFT}} X(k)$

WKT, $\text{DFT} \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

Let $x(n) = x(n+N)$, then

$$\text{DFT} \{x(n+N)\} = \sum_{n=0}^{N-1} x(n+N) W_N^{kn}$$

Put $d = n+N \Rightarrow n = d-N$

limits : $n=0$, $d=N$

$n=N-1$, $d=N-1+N = 2N-1$

$$= \sum_{l=N}^{2N-1} x(l) w_N^{k(l-N)}$$

$$= \sum_{l=N}^{2N-1} x(l) w_N^{kl} \cdot w_N^{-kN}$$

Since $w_N^{-kN} = 1$ & limits N to $2N-1$ can be replaced by 0 to $N-1$ using periodicity.

$$\text{DFT } \{ x(m+N) \} = \sum_{l=0}^{N-1} x(l) w_N^{kl}$$

$$\therefore \boxed{\text{DFT } \{ x(m+N) \} = X(k)}$$

Hence proved.

③ CIRCULAR TIME-SHIFT

Circular time shift operation on an N -point sequence $x(n)$ is given by $x((n-m))_{\text{mod } N}$, $x((n-m))_N$ or $x(n-m, \text{mod } N)$.

STATEMENT: If $x(n) \xleftrightarrow{\text{DFT}} X(k)$

$$\text{then } x((n-m))_N \xleftrightarrow{\text{DFT}} X(k) e^{-j\frac{2\pi}{N}km}$$

$$\text{or } x((n-m))_N \xleftrightarrow{\text{DFT}} X(k) W_N^{km}$$

$$\text{||| by } x((n+m))_N \xleftrightarrow{\text{DFT}} X(k) e^{j\frac{2\pi}{N}km}$$

$$\text{or } x((n+m))_N \xleftrightarrow{\text{DFT}} X(k) W_N^{-km}$$

Proof: W.K.T $X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

$$\text{Let } x(n) = x((n-m))_N$$

$$\text{then DFT } \{ x((n-m))_N \} = \sum_{n=0}^{N-1} x((n-m))_N \cdot W_N^{kn}$$

$$\text{Let } n-m = d \Rightarrow n = d+m, \text{ limits:}$$

$$\text{at } n=0, d = -m$$

$$n = N-1, d = N-1-m$$

$$\text{DFT } \{ x((n-m))_N \} = \sum_{d=-m}^{N-1-m} x(d) W_N^{k(d+m)}$$

$$= \sum_{d=0}^{N-1} x(d) W_N^{kd} \cdot W_N^{km} \quad \text{using periodicity}$$

$$\therefore \boxed{\text{DFT } \{ x((n-m))_N \} = X(k) W_N^{km}}$$

Example (15) If $x(n) = \{1, 2, 3, 4\}$ find $y(n)$,
given $y(n) = x((n-3))_4$.

Soln:- Given $y(n) = x((n-3))_4$

$$\begin{aligned} \text{at } n=0, \quad y(0) &= x((-3))_4 = x(4-3) \\ &= x(1) = \underline{\underline{2}} \end{aligned}$$

$$\begin{aligned} \text{at } n=1, \quad y(1) &= x((1-3))_4 = x((-2))_4 \\ &= x(4-2) = x(2) = \underline{\underline{3}} \end{aligned}$$

$$\begin{aligned} \text{at } n=2, \quad y(2) &= x((2-3))_4 = x((-1))_4 \\ &= x(4-1) = x(3) = \underline{\underline{4}} \end{aligned}$$

$$\begin{aligned} \text{at } n=3, \quad y(3) &= x((3-3))_4 = x(0) \\ &= \underline{\underline{1}} \end{aligned}$$

$$\therefore y(n) = \{2, 3, 4, 1\}$$

(16) If $x(n) = \{1, 2, 2, 1\}$ find the DFT of
 $y(n) = x((n-2))_4$.

Soln:- First find $X(k)$.

$$X_N = x_N W_N$$

$$X(k) = \begin{bmatrix} 1 & 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$\therefore X(k) = \begin{bmatrix} 6 & -1-j & 0 & -1+j \end{bmatrix}$$

Given $y(n) = x((n-2))_4$

Taking DFT and applying circular time-shift property,

$$Y(k) = X(k) W_4^{2k}, \quad k=0,1,2,3$$

at $k=0$, $Y(0) = X(0) W_4^0 = (6)(1) = 6$

at $k=1$, $Y(1) = X(1) W_4^2 = (-1-j)(-1) = 1+j$

at $k=2$, $Y(2) = X(2) W_4^4 = 0$

at $k=3$, $Y(3) = X(3) W_4^6 = (-1+j)(-1) = 1-j$

$$\therefore Y(k) = \{ 6, 1+j, 0, 1-j \}$$

4) TIME REVERSAL

STATEMENT: If $x(n) \xleftrightarrow[N]{\text{DFT}} X(k)$

then $x((-n))_N = x(N-n) \xleftrightarrow[N]{\text{DFT}}$

$$X((-k))_N = X(N-k)$$

Proof: - From def'n of DFT, we have

$$\text{DFT} \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$\text{Hence, DFT} \{x(N-n)\} = \sum_{n=0}^{N-1} x(N-n) W_N^{kn}$$

$$\text{Put } m = N-n \Rightarrow n = N-m$$

limits: at $n=0$, $m=N$

$$n=N-1, m=N-(N-1) = 1$$

$$\therefore \text{DFT} \{x(N-n)\} = \sum_{m=N}^1 x(m) W_N^{k(N-m)}$$

$$= \sum_{m=0}^{N-1} x(m) W_N^{kN} \cdot W_N^{-km} \quad \text{using periodicity property.}$$

$$\text{Also, } W_N^{kN} = 1$$

$$\text{DFT} \{x(N-n)\} = \sum_{m=0}^{N-1} x(m) W_N^{(-k)m}$$

$$\text{or } \boxed{\text{DFT} \{x(N-n)\} = X((-k))_N}$$

(17) Example : $x(n) = \{5, 2, -3, 7\}$. Find
 $y(n) = x((-n))_4$.

Sol'n: Consider $y(n) = x((-n))_4$
 or $y(n) = x(4-n)$

$$\text{at } n=0, \quad y(0) = x(4) = x(0) = 5$$

$$n=1, \quad y(1) = x(4-1) = x(3) = 7$$

$$n=2, \quad y(2) = x(4-2) = x(2) = -3$$

$$n=3, \quad y(3) = x(4-3) = x(1) = 2$$

$$\therefore y(n) = \{5, 7, -3, 2\}$$

(18) Given $x(n) = \{1, 2, 3, 4\}$ with 4-point
 DFT $X(k) = \{10, -2+2j, -2, -2-2j\}$. Find
 the 4-point DFT of the sequence
 $y(n) = \{1, 4, 3, 2\}$.

Sol'n:- Observing $x(n)$ & $y(n)$ we find that
 $y(n) = x((-n))_4$

Taking DFT on both sides and applying
 time reversal property

$$Y(k) = X((-k))_4 = X(4-k)$$

$$\text{at } k=0, \quad y(0) = x(0) = 10$$

$$k=1, \quad y(1) = x(4-1) = x(3) = -2-2j$$

$$k=2, \quad y(2) = x(4-2) = x(2) = -2$$

$$k=3, \quad y(3) = x(4-3) = x(1) = -2+2j$$

$$\therefore y(k) = \{ 10, -2-2j, -2, -2+2j \}$$

5) CIRCULAR FREQUENCY SHIFT

STATEMENT: $\mathcal{F}\{x(n)\} \xleftrightarrow[N]{\text{DFT}} X(k)$

then $x(n) e^{j\frac{2\pi}{N}mn} \xleftrightarrow[N]{\text{DFT}} X((k-m))_N$

or $x(n) W_N^{-mn} \xleftrightarrow{\text{DFT}} X((k-m))_N$

||| ly $x(n) e^{-j\frac{2\pi}{N}mn} \xleftrightarrow{\text{DFT}} X((k+m))_N$

or $x(n) W_N^{mn} \xleftrightarrow{\text{DFT}} X((k+m))_N$

Proof:- $X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

$$\text{DFT} \left\{ x(n) e^{j\frac{2\pi}{N}mn} \right\} = \sum_{n=0}^{N-1} x(n) e^{j\frac{2\pi}{N}mn} \cdot e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}(k-m)n}$$

$$= X((k-m))_N$$

Hence the proof.

(19) Find N-point DFT of $x_1(m) = \cos\left(\frac{2\pi}{N} k_0 m\right) x(m)$.

Soln:-

$$\text{Given } x_1(m) = \cos\left(\frac{2\pi}{N} k_0 m\right) x(m)$$

$$= \left[\frac{e^{j\frac{2\pi}{N} k_0 m} + e^{-j\frac{2\pi}{N} k_0 m}}{2} \right] x(m)$$

$$\text{If } x(m) \xleftrightarrow{\text{DFT}} X(k)$$

Using circular frequency shift property,

$$e^{j\frac{2\pi}{N} k_0 m} x(m) \xleftrightarrow{\text{DFT}} X((k - k_0))_N$$

$$e^{-j\frac{2\pi}{N} k_0 m} x(m) \xleftrightarrow{\text{DFT}} X((k + k_0))_N$$

$$\text{Hence DFT } \{x_1(m)\} = X_1(k)$$

$$\text{and } X_1(k) = \frac{1}{2} \left[X((k - k_0))_N + X((k + k_0))_N \right]$$

(20) Find N-point DFT of $x_2(m) = \sin\left(\frac{2\pi}{N} k_0 m\right) x(m)$

$$\underline{\text{ANS}} : X_2(k) = \frac{1}{2j} \left[X((k - k_0))_N - X((k + k_0))_N \right]$$

$$\text{Also, } x(m) \cos\left(\frac{4\pi}{N} m\right) \xleftrightarrow{\text{DFT}} \frac{1}{2} \left[X((k - 2))_N + X((k + 2))_N \right]$$

$$x(m) \sin\left(\frac{8\pi}{N} m\right) \xleftrightarrow{\text{DFT}} \frac{1}{2j} \left[X((k - 4))_N - X((k + 4))_N \right]$$

6) CIRCULAR CONVOLUTION / TIME-DOMAIN CONVOLUTION OR FREQUENCY DOMAIN MULTIPLICATION

STATEMENT:

$$\text{If } x(n) \xleftrightarrow{\text{DFT}} X(k)$$

$$\& h(n) \xleftrightarrow{\text{DFT}} H(k)$$

$$\text{then } y(n) = x(n) \circledast h(n) \xleftrightarrow{\text{DFT}} X(k) H(k) = Y(k)$$

Proof:

Consider two sequences $x(n)$ & $h(n)$ of length N , then circular convolution is defined as,

$$y(n) = x(n) \circledast h(n) = \sum_{m=0}^{N-1} x(m) h((n-m))_N$$

$$\text{DFT } \{y(n)\} = Y(k) = \sum_{n=0}^{N-1} y(n) W_N^{kn}$$

$$= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} x(m) h((n-m))_N \cdot W_N^{kn}$$

Interchanging order of summations,

$$= \sum_{m=0}^{N-1} x(m) \sum_{n=0}^{N-1} h((n-m))_N W_N^{kn}$$

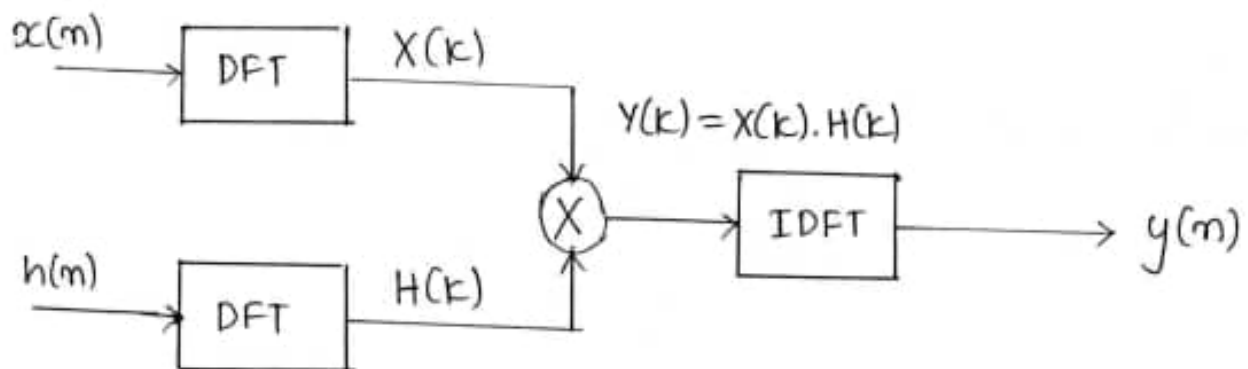
$$= \sum_{m=0}^{N-1} x(m) W_N^{km} H(k)$$

$$= X(k) H(k)$$

Using Time-shift property,
 $h((n-m))_N \xleftrightarrow{\text{DFT}} W_N^{km} H(k)$

Hence the proof.

II METHOD : Circular convolution using DFT & IDFT (Stockham's Method)



The method involves taking the N point DFTs of $x(m)$ & $h(m)$ both of length N points.

The respective DFTs are multiplied elementwise.

Then taking IDFT of the sequence $Y(k)$ to obtain $y(m)$.

- (21) Compute the circular convolution of sequences
 $x(m) = \{1, 2, 3, 4\}$ and $h(m) = \{1, 2, 2\}$.

Soln:- Given $x(m) = \{1, 2, 3, 4\}$

& $h(m) = \{1, 2, 2\}$

Length of $x(m)$, $N_1 = 4$. Length of $h(m)$, $N_2 = 3$.

The convolution length $N = \max(N_1, N_2)$
 $= \max(4, 3)$

$$\boxed{N = 4}$$

Since $h(m)$ is of length 3, pad one zero.

$$\therefore h(m) = \{1, 2, 2, 0\}$$

I-METHOD: Time-Domain Approach or
Concentric circle Method

From the def'n of circular convolution,

$$y(m) = x(m) \circledast h(m) = \sum_{n=0}^{N-1} x(m) h((m-n))_N$$

where $m=0$ to $N-1$.

Given $N=4$,

$$\therefore y(m) = \sum_{n=0}^3 x(m) h((m-n))_4, \quad m=0,1,2,3$$

$$= x(0) h((m))_4 + x(1) h((m-1))_4 \\ + x(2) h((m-2))_4 + x(3) h((m-3))_4$$

$$\text{at } m=0, \quad y(0) = x(0) h(0) + x(1) h((-1))_4 + x(2) h((-2))_4 \\ + x(3) h((-3))_4$$

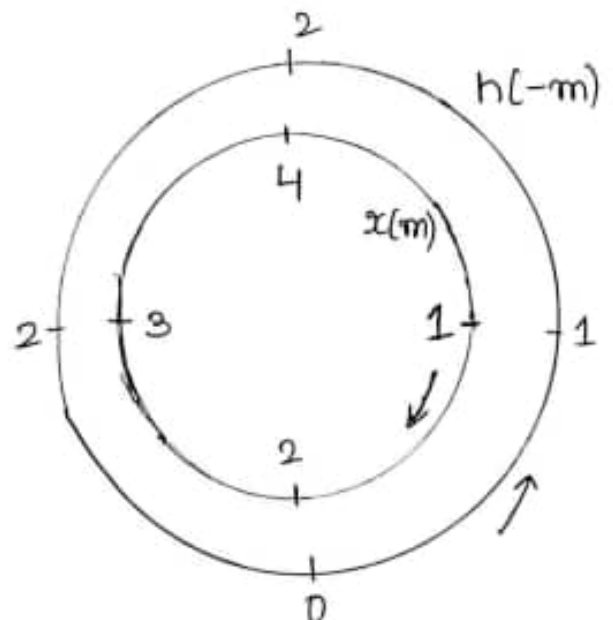
$$y(0) = x(0) h(0) + x(1) h(3)$$

$$+ x(2) h(2) + x(3) h(1)$$

$$= (1)(1) + (2)(0) + (3)(2) \\ + (4)(2)$$

$$= 1 + 0 + 6 + 8$$

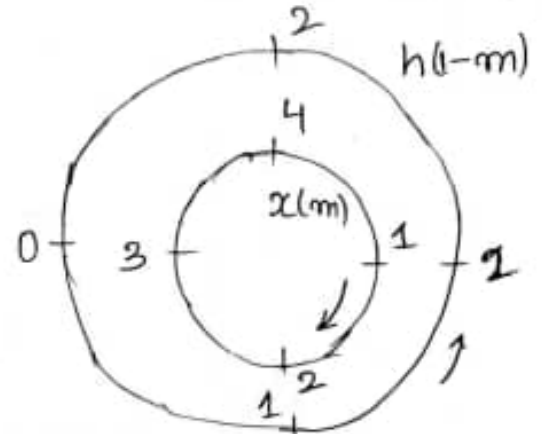
$$y(0) = \underline{\underline{15}}$$



at $n=1$, $y(1) = x(0)h(1) + x(1)h(0) + x(2)h(3) + x(3)h(2)$

$$y(1) = 2 + 2 + 0 + 8$$

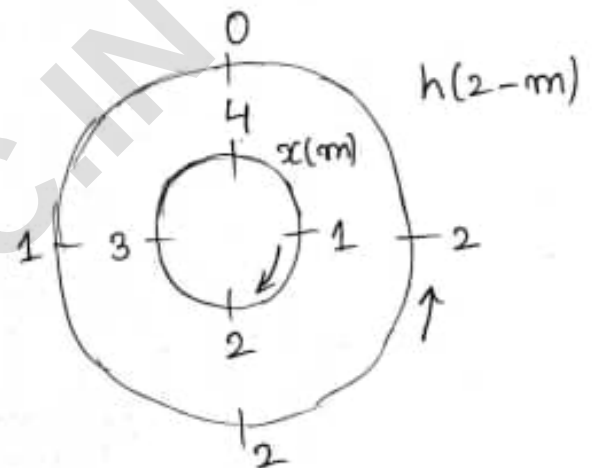
$$y(1) = \underline{\underline{12}}$$



at $n=2$, $y(2) = x(0)h(2) + x(1)h(1) + x(2)h(0) + x(3)h(3)$

$$y(2) = 2 + 4 + 3 + 0$$

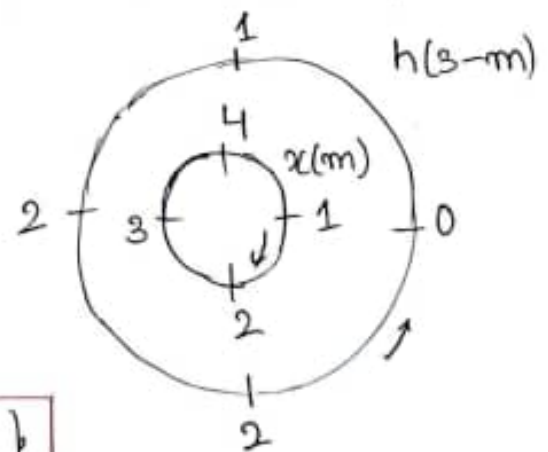
$$y(2) = \underline{\underline{9}}$$



at $n=3$, $y(3) = x(0)h(3) + x(1)h(2) + x(2)h(1) + x(3)h(0)$

$$y(3) = 0 + 4 + 6 + 4$$

$$y(3) = \underline{\underline{14}}$$



$$\therefore y(m) = \{ 15, 12, 9, 14 \}$$

Verification:

$$\begin{bmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \\ 9 \\ 14 \end{bmatrix}$$

II - METHOD: Using DFT & IDFT equations or Transform Domain Approach or Stockham's method

$$y(n) = x(n) \circledast h(n)$$

Applying DFT,

$$Y(k) = X(k) \cdot H(k)$$

(i) To find $X(k)$. $X_N = x_N W_N$

$$X(k) = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$X(k) = \{ 10, -2+2j, -2, -2-2j \}$$

(ii) To find $H(k)$. $H_N = h_N W_N$

$$H(k) = \begin{bmatrix} 1 & 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$H(k) = \{ 5, -1-2j, 1, -1+2j \}$$

(iii) Find the product of the sequence $x(k)$ & $H(k)$.

$$Y(k) = X(k) \cdot H(k)$$

$$Y(k) = \{ 50, 6+2j, -2, 6-2j \}$$

(iv) To find $y(m)$

$$y(m) = \text{IDFT} \{ Y(k) \} = \frac{1}{N} Y_N W_N^*$$

$$y(m) = \frac{1}{4} [50 \quad 6+2j \quad -2 \quad 6-2j] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

$$\therefore y(m) = \{ 15, 12, 9, 14 \}$$

7) MULTIPLICATION OF TWO SEQUENCES OR MODULATION PROPERTY OR CONVOLUTION IN FREQUENCY DOMAIN

STATEMENT: $\mathcal{F}\{x_1(n)\} \xleftrightarrow{\text{DFT}} X_1(k)$

& $x_2(n) \xleftrightarrow{\text{DFT}} X_2(k)$

then $x_1(n)x_2(n) \xleftrightarrow{\text{DFT}} \frac{1}{N} [X_1(k) \circledast X_2(k)]$

Proof:- Consider two sequences $x_1(n)$ & $x_2(n)$ of length N , with DFT $X_1(k)$ & $X_2(k)$.

$$\text{IDFT} \{X_1(k)\} = x_1(n) = \frac{1}{N} \sum_{l=0}^{N-1} X_1(l) W_N^{-ln}$$

$$\text{IDFT} \{X_2(k)\} = x_2(n) = \frac{1}{N} \sum_{m=0}^{N-1} X_2(m) W_N^{-mn}$$

Let $y(n) = x_1(n) \cdot x_2(n)$

then,

$$\begin{aligned} \text{DFT} \{y(n)\} &= Y(k) = \sum_{n=0}^{N-1} y(n) W_N^{kn} \\ &= \sum_{n=0}^{N-1} x_1(n) x_2(n) W_N^{kn} \end{aligned}$$

$$= \sum_{n=0}^{N-1} \left\{ \frac{1}{N} \sum_{l=0}^{N-1} X_1(l) W_N^{-ln} \right\} \left\{ \frac{1}{N} \sum_{m=0}^{N-1} X_2(m) W_N^{-mn} \right\} \\ \times W_N^{kn}$$

$$= \frac{1}{N^2} \sum_{l=0}^{N-1} X_1(l) \cdot \sum_{m=0}^{N-1} X_2(m) \sum_{n=0}^{N-1} W_N^{(k-l-m)n}$$

$$\text{If, } k-l-m = PN$$

$$\text{or } m = k-l-PN = ((k-l))_N$$

$$\text{or } l = k-m-PN = ((k-m))_N$$

$$\text{then, } \sum_{n=0}^{N-1} W_N^{(k-l-m)n} = \sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N} PNn} \\ = \sum_{n=0}^{N-1} 1 = N$$

$$\therefore Y(k) = \frac{1}{N^2} \sum_{l=0}^{N-1} X_1(l) X_2((k-l))_N \cdot N$$

$$Y(k) = \frac{1}{N} [X_1(k) \circledast X_2(k)]$$

8) SYMMETRY PROPERTY OF A COMPLEX VALUED SEQUENCE

STATEMENT: If $x(n) \xleftrightarrow{\text{DFT}} X(k)$

$$\text{then } x^*(n) \xleftrightarrow{\text{DFT}} X^*(N-k) = X^*((-k))_N$$

$$\text{or } x^*((-n))_N = x^*(N-n) \xleftrightarrow{\text{DFT}} X^*(k)$$

Proof: From def'n of DFT, we know that

$$\text{DFT } \{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

Taking complex conjugate on both sides,

$$X^*(k) = \sum_{n=0}^{N-1} x^*(n) W_N^{-kn}$$

replace k by $N-k$,

$$\begin{aligned} X^*(N-k) &= \sum_{n=0}^{N-1} x^*(n) W_N^{-(N-k)n} \\ &= \sum_{n=0}^{N-1} x^*(n) W_N^{-Nm} W_N^{km} \\ &= \sum_{n=0}^{N-1} x^*(n) W_N^{km} \quad \because W_N^{-Nm} = 1 \end{aligned}$$

$$X^*(N-k) = \text{DFT } \{x^*(n)\}$$

$$\text{or } \boxed{x^*(n) \xleftrightarrow[N]{\text{DFT}} X^*(N-k)}$$

9) SYMMETRY PROPERTY OF REAL-VALUED SEQUENCE

STATEMENT: If $x(n) \xleftrightarrow{\text{DFT}} X(k)$

and $x(n)$ is real, then

$$X(k) = X^*(N-k)$$

Proof:- W.K.T, $X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

Taking complex conjugate on both sides,

$$X^*(k) = \sum_{n=0}^{N-1} x^*(n) W_N^{-kn}$$

Since $x(n)$ is real, $x^*(n) = x(n)$

$$X^*(k) = \sum_{n=0}^{N-1} x(n) W_N^{-kn}$$

$$= \sum_{n=0}^{N-1} x(n) W_N^{-kn} \cdot W_N^{Nm} \quad \because W_N^{Nm} = 1$$

$$= \sum_{n=0}^{N-1} x(n) W_N^{(N-k)n}$$

$$X^*(k) = X(N-k)$$

$$\text{or } \boxed{X(k) = X^*(N-k)}$$

- ② The first five points of the 8-point DFT of a real-valued sequence are
 $\{ 0.25, (0.125 - j0.3018), 0, (0.125 - j0.0518), 0 \}$
 Determining the remaining 3 points.

Soln:- Given,

$$X(0) = 0.25$$

$$X(3) = 0.125 - j0.518$$

$$X(1) = 0.125 - j0.3018$$

$$X(4) = 0$$

$$X(2) = 0$$

Since $x(n)$ is real valued

$$X^*(k) = X(N-k)$$

$$\text{or } X(k) = X^*(N-k)$$

Here, $N=8$,

Then, for $k=5$, $X(5) = X^*(8-5) = X^*(3)$

$$= (0.125 - j0.0518)^*$$

$$X(5) = \underline{\underline{0.125 + j0.0518}}$$

for $k=6$, $X(6) = X^*(8-6) = X^*(2)$

$$= \underline{\underline{0}}$$

for $k=7$, $X(7) = X^*(8-7) = X^*(1)$

$$= (0.125 - j0.3018)^*$$

$$= \underline{\underline{0.125 + j0.3018}}$$

10) CIRCULAR CORRELATION

For complex valued sequences $x(n)$ and $y(n)$

$$\text{If } x(n) \xleftrightarrow{\text{DFT}} X(k)$$

$$y(n) \xleftrightarrow{\text{DFT}} Y(k)$$

$$\text{then } r_{xy}(l) \xleftrightarrow{\text{DFT}} R_{xy}(k) = X(k) \cdot Y^*(k)$$

where $r_{xy}(l)$ is the circular cross-correlation sequence defined by

$$r_{xy}(l) = \sum_{n=0}^{N-1} x(n) y^*((n-l))_N$$

Proof: We can express $r_{xy}(l)$ as the circular convolution of $x(n)$ with $y^*(-n)$.

$$\text{i.e., } r_{xy}(l) = x(l) \circledast y^*(-l)$$

taking DFT on both sides,

$$\text{DFT } \{ r_{xy}(l) \} = \text{DFT } \{ x(l) \circledast y^*(-l) \}$$

$$\boxed{R_{xy}(k) = X(k) \cdot Y^*(k)}$$

using circular convolution & complex conjugate property

$$\text{If } x(n) = y(n),$$

$$r_{xx}(l) \xleftrightarrow{\text{DFT}} R_{xx}(k) = |X(k)|^2$$

11) PARSEVAL'S THEOREM

For complex valued sequence $x(n)$ & $y(n)$

$$\text{If } x(n) \xleftrightarrow{\text{DFT}} X(k)$$

$$y(n) \xleftrightarrow{\text{DFT}} Y(k)$$

$$\text{then } \sum_{n=0}^{N-1} x(n) \cdot y^*(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) \cdot Y^*(k)$$

Proof:- Circular cross correlation of two sequences is defined as

$$r_{xy}(l) = \sum_{n=0}^{N-1} x(n) y^*((n-l))_N$$

$$\text{at } l=0, \quad r_{xy}(0) = \sum_{n=0}^{N-1} x(n) y^*(n) \quad \text{--- (a)}$$

$$\text{W.K.T, } \text{IDFT} \{ R_{xy}(k) \} = r_{xy}(l)$$

$$\text{or } r_{xy}(l) = \frac{1}{N} \sum_{k=0}^{N-1} R_{xy}(k) e^{j\frac{2\pi}{N}kl}$$

from def'n of IDFT

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k) e^{j\frac{2\pi}{N}kl}$$

from circular correlation property

at $l=0$,

$$r_{xy}(0) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k) \quad \text{--- (b)}$$

Equating RHS of Eqs (a) & (b),

$$\sum_{n=0}^{N-1} x(n) y^*(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k)$$

If $x(n) = y(n)$, then

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

which is the energy in the finite duration sequence $x(n)$ in terms of the frequency components $X(k)$.

- (23) Determine N-point circular correlation of $x(n) = \cos \frac{2\pi n}{N}$ & $y(n) = \sin \frac{2\pi n}{N}$.

Soln:- Given $x(n) = \cos \frac{2\pi n}{N}$

Taking N-point DFT,

$$X(k) = \frac{N}{2} [\delta(k-1) + \delta(k+1)]$$

Given $y(n) = \sin \frac{2\pi n}{N}$

Its N-point DFT, $Y(k) = \frac{N}{2j} [\delta(k-1) - \delta(k+1)]$

From circular correlation property,

$$R_{xy}(k) = X(k) \cdot Y^*(k)$$

$$R_{xy}(k) = \frac{N}{2} [\delta(k-1) + \delta(k+1)] \cdot \left\{ \frac{N}{2j} [\delta(k-1) - \delta(k+1)] \right\}^*$$

$$= \frac{N}{2} [\delta(k-1) + \delta(k+1)] \left\{ -\frac{N}{2j} [\delta(k-1) - \delta(k+1)] \right\}$$

$$= -\frac{N^2}{4j} \left\{ [\delta(k-1)]^2 + \delta(k+1)\delta(k-1) - \delta(k-1)\delta(k+1) - [\delta(k+1)]^2 \right\}$$

WKT, $\delta(k+1)\delta(k-1) = 0$, $[\delta(k-1)]^2 = \delta(k-1)$
and $[\delta(k+1)]^2 = \delta(k+1)$.

$$= -\frac{N^2}{4j} [\delta(k-1) - \delta(k+1)]$$

$$R_{xy}(k) = -\frac{N}{2} \left\{ \frac{N}{2j} [\delta(k-1) - \delta(k+1)] \right\}$$

taking IDFT,

$$r_{xy}(l) = -\frac{N}{2} \sin \frac{2\pi}{N} n$$

(24) Find the circular autocorrelation of $x(n) = \cos \frac{2\pi}{N}n$

Soln:— Given $x(n) = \cos\left(\frac{2\pi}{N}n\right)$

Taking DFT,

$$X(k) = \frac{N}{2} [\delta(k-1) + \delta(k+1)]$$

WKT $R_{xy}(k) = X(k) \cdot Y^*(k)$

Since $x(n) = y(n)$,

$$R_{xx}(k) = X(k) \cdot X^*(k)$$

$$R_{xx}(k) = \left\{ \frac{N}{2} [\delta(k-1) + \delta(k+1)] \right\} \left\{ \frac{N}{2} [\delta(k-1) + \delta(k+1)] \right\}^*$$

$$= \frac{N}{2} \cdot \frac{N}{2} [\delta(k-1) + \delta(k+1)]$$

taking IDFT,

$$\boxed{r_{xx}(n) = \frac{N}{2} \cos\left(\frac{2\pi}{N}n\right)}$$

- (25) Find circular autocorrelation of the sequence $x(m) = \{1, 2, 3, 4\}$.

Soln:- Given $x(m) = \{1, 2, 3, 4\}$, $N=4$

Find the 4-point DFT,

$$\text{we get } X(k) = \{10, -2+2j, -2, -2-2j\}$$

Autocorrelation of the sequence $x(m)$ is

$$\tilde{r}_{xx}(l) \xleftrightarrow{\text{DFT}} \tilde{R}_{xx}(k) = X(k) \cdot X^*(k) \\ = |X(k)|^2$$

$$\tilde{R}_{xx}(0) = |X(0)|^2 = 10^2 = 100$$

$$\tilde{R}_{xx}(1) = |X(1)|^2 = |-2+2j|^2 \\ = |\sqrt{4+4}|^2 = 8$$

$$\tilde{R}_{xx}(2) = |X(2)|^2 = |-2|^2 = 4$$

$$\tilde{R}_{xx}(3) = |X(3)|^2 = |-2-2j|^2 = 8$$

$$\therefore \boxed{\tilde{R}_{xx}(k) = \{100, 8, 4, 8\}}$$

To find $\tilde{r}_{xx}(l)$:

$$\tilde{r}_{xx}(l) = \text{IDFT} \{ R_{xx}(k) \}$$

$$\begin{aligned} \tilde{r}_{xx}(l) &= \frac{1}{4} \begin{bmatrix} 100 & 8 & 4 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 120 & 96 & 88 & 96 \end{bmatrix} \end{aligned}$$

$$\tilde{r}_{xx}(l) = \{ 30, 24, 22, 24 \}$$

(26) Find the circular correlation given

$$x(n) = \{ 1, 2, 3, 4 \} \text{ and } y(n) = \{ 1, 2, 2, 0 \}$$

Soln:- Find $X(k)$

$$X(k) = \{ 10, -2+2j, -2, -2-2j \}$$

Find $Y(k)$

$$Y(k) = \{ 5, -1-2j, 1, -1+2j \}$$

$$\Rightarrow Y^*(k) = \{ 5, -1+2j, 1, -1-2j \}$$

Circular correlation property,

$$R_{xy}(k) = X(k) \cdot Y^*(k)$$

$$= \{ 50, -2-6j, -2, -2+6j \}$$

Taking IDFT,

$$r_{xy}(l) = \{ 11, 16, 13, 10 \}$$

(27) Given $x(n) = \{ 1, 2, 3, 4 \}$ find the energy and hence verify Parseval's theorem.

Soln:- Given $x(n) = \{ 1, 2, 3, 4 \}$

Energy of the signal $x(n)$ is given by

$$E = \sum_{n=-\infty}^{+\infty} |x(n)|^2$$

$$= \sum_{n=0}^3 |x(n)|^2 = |x(0)|^2 + |x(1)|^2 + |x(2)|^2 + |x(3)|^2$$

$$= 1^2 + 2^2 + 3^2 + 4^2$$

$$E = 30$$

Find DFT of $x(n)$,

$$X(k) = \{ 10, -2+2j, -2, -2-2j \}$$

From Parseval's Theorem,

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

$$E = \frac{1}{4} \sum_{k=0}^3 |X(k)|^2$$

$$= \frac{1}{4} \left[|X(0)|^2 + |X(1)|^2 + |X(2)|^2 + |X(3)|^2 \right]$$

$$= \frac{1}{4} [100 + 8 + 4 + 8]$$

$$E = \frac{120}{4} = \underline{\underline{30}}$$

Hence Parseval's Theorem is verified.

Definition of Even and Odd Symmetry and time reversal.

An N -point sequence is said to be circularly even if it is symmetric about the point zero on the circle.

$$\text{ie, } \boxed{x(N-m) = x(m)}, \quad 1 \leq m \leq N-1$$

An N -point sequence is said to be circularly odd if it is antisymmetric about the point zero on the circle.

$$\text{ie, } \boxed{x(N-m) = -x(m)}, \quad 1 \leq m \leq N-1$$

The time reversal of an N -point sequence is attained by reversing its samples about the point zero on the circle. Thus the sequence $x((-m))_N$ is given by

$$\boxed{x((-m))_N = x(N-m)}, \quad 0 \leq m \leq N-1$$

For periodic sequence $x_p(m)$, def'n of even and odd sequences :

$$\begin{aligned} \text{even: } & \boxed{x_p(m) = x_p(-m) = x_p(N-m)} \\ \text{odd: } & \boxed{x_p(m) = -x_p(-m) = -x_p(N-m)} \end{aligned}$$

If $x_p(m)$ is complex valued, then

$$\begin{aligned} \text{conjugate even: } & \boxed{x_p(m) = x_p^*(N-m)} \\ \text{conjugate odd: } & \boxed{x_p(m) = -x_p^*(N-m)} \end{aligned}$$

Sequence $x_p(m)$ can be decomposed as

$$x_p(m) = x_{pe}(m) + x_{po}(m)$$

where

$$\begin{aligned} x_{pe}(m) &= \frac{1}{2} [x_p(m) + x_p^*(N-m)] \\ x_{po}(m) &= \frac{1}{2} [x_p(m) - x_p^*(N-m)] \end{aligned}$$

Symmetry properties of the DFT

Let us assume that $x(n)$ and its DFT $X(k)$ are both complex valued.

$$\text{Then, } x(n) = x_R(n) + j x_I(n), \quad n=0, 1, \dots, N-1 \quad \text{--- (1)}$$

$$X(k) = X_R(k) + j X_I(k), \quad k=0, 1, \dots, N-1 \quad \text{--- (2)}$$

From the def'n of DFT,

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad \text{--- (3)}$$

substituting eqn (1) in eqn (3),

$$X(k) = \sum_{n=0}^{N-1} [x_R(n) + j x_I(n)] e^{-j \frac{2\pi}{N} kn}$$

and since, $e^{j\theta} = \cos\theta + j \sin\theta$

$$\begin{aligned}
 X(k) &= \sum_{n=0}^{N-1} \left[x_R(n) + j x_I(n) \right] \left[\cos \frac{2\pi}{N} kn - j \sin \frac{2\pi}{N} kn \right] \\
 &= \sum_{n=0}^{N-1} \left[x_R(n) \cos \left(\frac{2\pi}{N} kn \right) - j x_R(n) \sin \left(\frac{2\pi}{N} kn \right) \right. \\
 &\quad \left. + j x_I(n) \cos \left(\frac{2\pi}{N} kn \right) + x_I(n) \sin \left(\frac{2\pi}{N} kn \right) \right]
 \end{aligned}
 \quad \text{--- (4)}$$

Comparing eqns (2) & (4),

$$\begin{aligned}
 X_R(k) &= \sum_{n=0}^{N-1} \left[x_R(n) \cos \left(\frac{2\pi}{N} kn \right) + x_I(n) \sin \left(\frac{2\pi}{N} kn \right) \right] \\
 X_I(k) &= \sum_{n=0}^{N-1} \left[-x_R(n) \sin \left(\frac{2\pi}{N} kn \right) + x_I(n) \cos \left(\frac{2\pi}{N} kn \right) \right]
 \end{aligned}
 \quad \begin{array}{l} \text{--- (5)} \\ \text{--- (6)} \end{array}$$

IIIrd, WKT $x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn}$

substituting eqn (2) in above equation,

$$\begin{aligned}
 x(n) &= \frac{1}{N} \sum_{k=0}^{N-1} \left[X_R(k) + j X_I(k) \right] e^{j \frac{2\pi}{N} kn} \\
 &= \frac{1}{N} \sum_{k=0}^{N-1} \left[X_R(k) + j X_I(k) \right] \left[\cos \left(\frac{2\pi}{N} kn \right) + j \sin \left(\frac{2\pi}{N} kn \right) \right]
 \end{aligned}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} \left[X_R(k) \cos\left(\frac{2\pi}{N} km\right) + j X_R(k) \sin\left(\frac{2\pi}{N} km\right) + j X_I(k) \cos\left(\frac{2\pi}{N} km\right) - X_I(k) \sin\left(\frac{2\pi}{N} km\right) \right]$$

↳ (7)

Comparing eqn's (1) & (7),

$$x_R(m) = \frac{1}{N} \sum_{k=0}^{N-1} \left[X_R(k) \cos\left(\frac{2\pi}{N} km\right) - X_I(k) \sin\left(\frac{2\pi}{N} km\right) \right]$$

↳ (8)

$$x_I(m) = \frac{1}{N} \sum_{k=0}^{N-1} \left[X_R(k) \sin\left(\frac{2\pi}{N} km\right) + X_I(k) \cos\left(\frac{2\pi}{N} km\right) \right]$$

↳ (9)

case 1) Real Valued sequences

if $x(m)$ is real, then

$$X(N-k) = X^*(k) = X(-k)$$

And, $|X(N-k)| = |X(k)|$ also $\angle X(N-k) = -\angle X(k)$

For real $x(m)$, $x_I(m) = 0$, hence $x(m)$ can be determined from eqn (8).

Case 2) Real and even sequence

If $x(n)$ is real and even, that is

$$x(n) = x(N-n), \quad 0 \leq n \leq N-1$$

then eqn. ⑥ yields $X_I(k) = 0$. Hence the DFT reduces to

$$X(k) = \sum_{n=0}^{N-1} x(n) \cos\left(\frac{2\pi kn}{N}\right), \quad 0 \leq k \leq N-1$$

where $X(k)$ is real valued and even.

Since $X_I(k) = 0$, the IDFT reduces to

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) \cos\left(\frac{2\pi kn}{N}\right), \quad 0 \leq n \leq N-1$$

Case 3) Real and odd sequence

If $x(n)$ is real and odd, that is

$$x(n) = -x(N-n), \quad 0 \leq n \leq N-1$$

then eqn. ⑤ yields $X_R(k) = 0$. Hence

$$X(k) = -j \sum_{n=0}^{N-1} x(n) \sin\left(\frac{2\pi kn}{N}\right), \quad 0 \leq k \leq N-1$$

which is purely imaginary and odd.

Since $X_R(k) = 0$, the IDFT reduces to

$$x(n) = j \frac{1}{N} \sum_{k=0}^{N-1} X(k) \sin\left(\frac{2\pi kn}{N}\right), \quad 0 \leq n \leq N-1$$

Case 4) Purely Imaginary sequence

In this case $x(n) = j x_I(n)$.

Eqn's (5) & (6) reduces to

$$X_R(k) = \sum_{n=0}^{N-1} x_I(n) \sin \frac{2\pi kn}{N}$$

$$X_I(k) = \sum_{n=0}^{N-1} x_I(n) \cos \frac{2\pi kn}{N}$$

We observe that $X_R(k)$ is odd and $X_I(k)$ is even.

If $x_I(n)$ is odd, then $X_I(k) = 0$, hence $X(k)$ is purely real.

If $x_I(n)$ is even, then $X_R(k) = 0$, hence $X(k)$ is purely imaginary.

The symmetry properties above may be summarised as follows:

$$\begin{array}{l} x(n) = x_R^e(n) + x_R^o(n) + j x_I^e(n) + j x_I^o(n) \\ X(k) = X_R^e(k) + X_R^o(k) + j X_I^e(k) + j X_I^o(k) \end{array}$$

PROPERTIES OF THE DFT

PROPERTY	TIME DOMAIN	FREQUENCY DOMAIN
Periodicity	$x(m) = x(m+N)$	$X(k) = X(k+N)$
Linearity	$a_1 x_1(m) + a_2 x_2(m)$	$a_1 X_1(k) + a_2 X_2(k)$
Time Reversal	$x(N-m)$	$X(N-k)$
Circular Time Shift	$x((m-l))_N$	$X(k) \cdot e^{-j\frac{2\pi k l}{N}}$
Circular Frequency Shift	$x(m) e^{j\frac{2\pi l m}{N}}$	$X((k-l))_N$
Complex Conjugate	$x^*(m)$	$X^*(N-k)$
Circular Convolution	$x_1(m) \circledast x_2(m)$	$X_1(k) X_2(k)$
Circular Correlation	$x(m) \circledast y^*(-m)$	$X(k) Y^*(k)$
Multiplication of two sequences	$x_1(m) x_2(m)$	$\frac{1}{N} [X_1(k) \circledast X_2(k)]$
Parseval's Theorem	$\sum_{n=0}^{N-1} x(m) y^*(m)$	$\frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k)$

- (28) A 498 point DFT of a real valued sequence $x(n)$ has the following DFT samples

$$X(0) = 2$$

$$X(249) = 2.9$$

$$X(11) = 7 + j3.1$$

$$X(309) = -4.7 - j1.9$$

$$X(k_1) = -2.2 - j1.5$$

$$X(k_3) = 3 - j0.7$$

$$X(112) = 3 + j0.7$$

$$X(412) = -2.2 + j1.5$$

$$X(k_2) = -4.7 + j1.9$$

$$X(k_4) = 7 - j3.1$$

The other samples have a value zero. Find the value of k_1, k_2, k_3 & k_4 .

Sol'n:- For real valued sequence,

$$\text{WKT, } X(k) = X^*(N-k)$$

Given $N = 498$.

$$\begin{aligned} \text{a) For } k = 412, \quad X(412) &= X^*(498 - 412) \\ &= X^*(86) \end{aligned}$$

$$\begin{aligned} \text{or } X(86) &= X^*(412) \\ &= (-2.2 + j1.5)^* \end{aligned}$$

$$X(86) = -2.2 - j1.5 = X(k_1)$$

$$\therefore \boxed{k_1 = 86}$$

$$\begin{aligned} \text{b) for } k=309, \quad X(309) &= X^*(498-309) \\ &= X^*(189) \end{aligned}$$

$$\begin{aligned} \text{or } X(189) &= X^*(309) \\ &= (-4.7 - j1.9)^* \end{aligned}$$

$$X(189) = -4.7 + j1.9 = X(k_2)$$

$$\therefore \boxed{k_2 = 189}$$

$$\begin{aligned} \text{c) for } k=112, \quad X(112) &= X^*(498-112) \\ &= X^*(386) \end{aligned}$$

$$\begin{aligned} \text{or } X(386) &= X^*(112) \\ &= (3 + j0.7)^* \end{aligned}$$

$$X(386) = 3 - j0.7 = X(k_3)$$

$$\therefore \boxed{k_3 = 386}$$

$$\begin{aligned} \text{d) for } k=11, \quad X(11) &= X^*(498-11) \\ &= X^*(487) \end{aligned}$$

$$\begin{aligned} \text{or } X(487) &= X^*(11) \\ &= (7 + j3.1)^* \end{aligned}$$

$$X(487) = 7 - j3.1 = X(k_4)$$

$$\therefore \boxed{k_4 = 487}$$

Hence, $k_1 = 86$, $k_2 = 386$, $k_3 = 189$ & $k_4 = 487$

(29) Find the 4-point DFTs of the two sequences $x(n)$ & $y(n)$ using a single 4-point DFT.
 $x(n) = \{1, 2, 0, 1\}$ & $y(n) = \{2, 2, 1, 1\}$.

Sol'n:-

From the given, combining the two sequences $x(n)$ and $y(n)$ to create a complex sequence $h(n)$.

$$\text{With, } h(n) = x(n) + jy(n), \quad 0 \leq n \leq 3$$

$$\therefore h(n) = \{(1+2j), (2+2j), j, (1+j)\}$$

$$\text{Taking DFT, } H_N = h_N W_N$$

$$\therefore H(k) = \begin{bmatrix} 1+2j & 2+2j & j & 1+j \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$H(k) = \begin{bmatrix} 4+6j & 2 & -2 & 2j \end{bmatrix}$$

$$\Rightarrow H^*(k) = \begin{bmatrix} 4-6j & 2 & -2 & -2j \end{bmatrix}$$

$$\text{Using the relation, } X(k) = \frac{H(k) + H^*((-k))_N}{2}$$

$$\begin{aligned} \text{for } k=0, \quad X(0) &= \frac{H(0) + H^*(0)}{2} = \frac{(4+6j) + (4-6j)}{2} \\ &= \frac{8}{2} = \underline{\underline{4}} \end{aligned}$$

$$\begin{aligned} \text{for } k=1, \quad X(1) &= \frac{H(1) + H^*((-1))_4}{2} = \frac{H(1) + H^*(3)}{2} \\ &= \frac{2 - 2j}{2} = \underline{\underline{1-j}} \end{aligned}$$

$$\begin{aligned} \text{for } k=2, \quad X(2) &= \frac{H(2) + H^*((-2))_4}{2} = \frac{H(2) + H^*(2)}{2} \\ &= \frac{-2 - 2}{2} = \underline{\underline{-2}} \end{aligned}$$

$$\begin{aligned} \text{for } k=3, \quad X(3) &= \frac{H(3) + H^*((-3))_4}{2} = \frac{H(3) + H^*(1)}{2} \\ &= \frac{2j + 2}{2} = \underline{\underline{1+j}} \end{aligned}$$

$$\therefore X(k) = \{ 4, (1-j), -2, (1+j) \}$$

$$\text{Also, } Y(k) = \frac{H(k) - H^*((-k))_4}{2j}$$

$$\text{for } k=0, \quad Y(0) = \frac{H(0) - H^*(0)}{2j} = \frac{4+6j - 4+6j}{2j} = \underline{\underline{6}}$$

$$\begin{aligned} \text{for } k=1, \quad Y(1) &= \frac{H(1) - H^*((-1))_4}{2j} = \frac{H(1) - H^*(3)}{2j} \\ &= \frac{2+2j}{2j} = \underline{\underline{1-j}} \end{aligned}$$

$$\begin{aligned} \text{for } k=2, \quad Y(2) &= \frac{H(2) - H^*((-2))_4}{2j} = \frac{H(2) - H^*(2)}{2j} \\ &= \frac{-2 + 2}{2j} = \underline{\underline{0}} \end{aligned}$$

$$\begin{aligned} \text{for } k=3, \quad Y(3) &= \frac{H(3) - H^*((-3))_4}{2j} = \frac{H(3) - H^*(1)}{2j} \\ &= \frac{2j - 2}{2j} = \underline{\underline{1+j}} \end{aligned}$$

$$\therefore Y(k) = \{ 6, (1-j), 0, (1+j) \}$$

- ③⑦ Let $x_p(m)$ be a periodic sequence with fundamental period N . If the N -point DFT $\{x_p(m)\} = X_1(k)$ and $3N$ -point DFT $\{x_p(m)\} = X_3(k)$.
- (i) Find the relationship between $X_1(k)$ & $X_3(k)$.
- (ii) Verify the above result for $\{2, 1\}$ & $\{2, 1, 2, 1, 2, 1\}$.

sol'n:- (i) WKT N -point DFT $\{x(m)\} = X(k)$

$$= \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$N\text{-point DFT } \{x_p(m)\} = X_1(k) = \sum_{n=0}^{N-1} x_p(n) W_N^{kn}$$

$$3N\text{-point DFT } \{x_p(m)\} = X_3(k) = \sum_{n=0}^{3N-1} x_p(n) W_{3N}^{kn}$$

$$X_3(k) = \sum_{n=0}^{N-1} x_p(n) W_{3N}^{kn} + \sum_{n=0}^{N-1} x_p(n+N) W_{3N}^{k(n+N)} + \sum_{n=0}^{N-1} x_p(n+2N) W_{3N}^{k(n+2N)}$$

$$= \sum_{n=0}^{N-1} x_p(n) W_{3N}^{kn} + \sum_{n=0}^{N-1} x_p(n) W_{3N}^{kn} \cdot W_{3N}^{kN}$$

$$+ \sum_{n=0}^{N-1} x_p(n) W_{3N}^{kn} \cdot W_{3N}^{2kN} \quad \because \text{From periodicity of input.}$$

$$x_p(n+N) = x_p(n)$$

$$\& x_p(n+2N) = x_p(n)$$

Also, $W_{3N}^{NK} = e^{-j\frac{2\pi}{3N}NK} = e^{-j\frac{2\pi}{3}K} = W_3^K$

Similarly $W_{3N}^{2NK} = W_3^{2K}$ & $W_{3N}^{mK} = W_N^{\frac{mK}{3}}$.

$$\therefore X_3(K) = \sum_{n=0}^{N-1} x_p(n) \left[1 + W_3^K + W_3^{2K} \right] W_N^{\frac{mK}{3}}$$

$$= \left[1 + W_3^K + W_3^{2K} \right] \sum_{n=0}^{N-1} x_p(n) W_N^{\frac{mK}{3}}$$

$$X_3(K) = \left[1 + W_3^K + W_3^{2K} \right] X_1(K/3)$$

ii) Given $x_1(m) = \{2, 1\}$

Its DFT, $X_1(K) = 2 + W_2^K$

$x_3(m) = \{2, 1, 2, 1, 2, 1\}$ with DFT

$$X_3(K) = 2 + W_6^K + 2W_6^{2K} + W_6^{3K} + 2W_6^{4K} + W_6^{5K}$$

$$W_6^K = W_{2 \times 3}^K = W_2^{K/3}$$

$$\text{Then, } X_3(K) = \left[2 + W_2^{K/3} \right] + W_6^{2K} \left[2 + W_2^{K/3} \right]$$

$$+ W_6^{4K} \left[2 + W_2^{K/3} \right]$$

$$= \left[2 + W_2^{K/3} \right] \left[1 + W_6^{2K} + W_6^{4K} \right]$$

$$X_3(K) = X_1(K/3) \left[1 + W_3^K + W_3^{2K} \right]$$

Hence verified [Studied smart not hard — thanks to VTUSync.in](https://studiedsmart.com)

- 31) Suppose that you are given a program to find the DFT of a complex-valued sequence $x(m)$. How can we use this program to find the inverse DFT of $X(k)$.

Sol'n:- WKT,

$$\text{DFT } \{x(m)\} = X(k) = \sum_{n=0}^{N-1} x(m) W_N^{km} \quad \text{--- (1)}$$

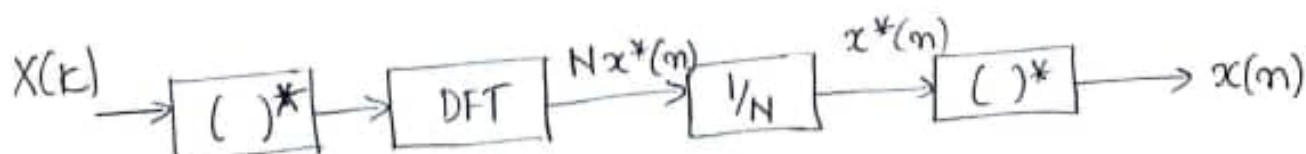
$$\text{IDFT } \{X(k)\} = x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-km} \quad \text{--- (2)}$$

Taking complex conjugate of eqn (2),

$$x^*(m) = \frac{1}{N} \sum_{k=0}^{N-1} X^*(k) W_N^{km}$$

$$\text{or } N x^*(m) = \sum_{k=0}^{N-1} X^*(k) W_N^{km}$$

From observing the above eqn, we can use DFT program to compute IDFT by calculating $X^*(k)$ and finding its DFT. The output is $N x^*(m)$. scale it by $\frac{1}{N}$ and take its conjugate.



- (32) Consider the sequence $x(n) = 4\delta(n) + 3\delta(n-1) + 2\delta(n-2) + \delta(n-3)$.

Let $X(k)$ be the 6-point DFT of $x(n)$. Find the finite length sequence $y(n)$ that has a 6-point DFT $Y(k) = W_6^{4k} X(k)$.

Soln:- Given, $x(n) = 4\delta(n) + 3\delta(n-1) + 2\delta(n-2) + \delta(n-3)$

Its DFT,
$$X(k) = 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}$$

Also, given $Y(k) = W_6^{4k} X(k)$

$$\begin{aligned} Y(k) &= W_6^{4k} [4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}] \\ &= 4W_6^{4k} + 3W_6^{5k} + 2W_6^{6k} + W_6^{7k} \end{aligned}$$

WKT, $W_6^{6k} = W_6^0 = 1$, $W_6^{7k} = W_6^k$

$$\therefore Y(k) = 4W_6^{4k} + 3W_6^{5k} + 2 + W_6^k$$

$$Y(k) = 2 + W_6^k + 4W_6^{4k} + 3W_6^{5k}$$

Taking IDFT,

$$y(n) = 2\delta(n) + \delta(n-1) + 4\delta(n-4) + 3\delta(n-5)$$

$$\therefore y(n) = \{2, 1, 0, 0, 4, 3\}$$

(33) If $x(n) = \{1, 2, 0, 3, -2, 4, 7, 5\}$.

Evaluate the following (i) $X(0)$ (ii) $X(4)$ (iii) $\sum_{k=0}^7 X(k)$
 (iv) $\sum_{k=0}^7 |X(k)|^2$

Sol'n:- (i) From the def'n of DFT we have

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$N=8,$

at $k=0$, $X(0) = \sum_{n=0}^7 x(n) W_8^0$

$$= \sum_{n=0}^7 x(n) = 1 + 2 + 0 + 3 - 2 + 4 + 7 + 5$$

$$\boxed{X(0) = 20}$$

(ii) $X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

for $N=8$, Let $k=4$

$$X(4) = \sum_{n=0}^7 x(n) W_8^{4n}$$

$$= \sum_{n=0}^7 x(n) (-1)^n$$

$$\begin{aligned} \because W_8^{4n} &= e^{j\frac{2\pi}{8} \times 4n} \\ &= (e^{-j\pi})^n = (-1)^n \end{aligned}$$

$$\begin{aligned} X(4) &= x(0) - x(1) + x(2) - x(3) + x(4) - x(5) \\ &\quad + x(6) - x(7) \end{aligned}$$

$$\boxed{X(4) = -8}$$

iii) To find $\sum_{k=0}^7 X(k)$

Consider the IDFT equation,

$$x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-km}$$

for $N=8$, put $m=0$,

$$x(0) = \frac{1}{8} \sum_{k=0}^7 X(k)$$

$$\therefore \sum_{k=0}^7 X(k) = 8 x(0) = \underline{\underline{8}}$$

iv) To find $\sum_{k=0}^7 |X(k)|^2$

From Parseval's Theorem,

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

$$\text{or } \sum_{k=0}^{N-1} |X(k)|^2 = N \sum_{n=0}^{N-1} |x(n)|^2$$

$$\text{for } N=8, \sum_{k=0}^7 |X(k)|^2 = 8 \left[\sum_{n=0}^7 |x(n)|^2 \right]$$

$$= 8 \left[|1|^2 + |2|^2 + |0|^2 + |3|^2 + |-2|^2 + |4|^2 + |7|^2 + |5|^2 \right]$$

$$= 8 [1 + 4 + 9 + 4 + 16 + 49 + 25]$$

$$\sum_{k=0}^7 |X(k)|^2 = \underline{\underline{864}}$$

34) Let $x(n)$ be a finite length sequence with $X(k) = \{0, (1+j), 1, (1-j)\}$ using the properties of DFT, find DFTs of the following sequences.

(i) $x_1(n) = e^{j\frac{\pi}{2}n} x(n)$

(ii) $x_2(n) = \cos\frac{\pi}{2}n x(n)$

(iii) $x_3(n) = x((n-1))_4$

Sol'n:- (i) Given $x_1(n) = e^{j\frac{\pi}{2}n} x(n)$

or $x_1(n) = e^{j\frac{2\pi}{4}n} x(n)$

Applying DFT and using circular frequency shift property

$$X_1(k) = X((k-1))_4$$

$$\therefore X_1(k) = \{ \underline{(1-j)}, 0, (1+j), 1 \}$$

(ii) Given $x_2(n) = \cos\frac{\pi}{2}n x(n)$

$$\text{or } x_2(n) = \frac{1}{2} \left[e^{j\frac{\pi}{2}n} + e^{-j\frac{\pi}{2}n} \right] x(n)$$

$$= \frac{1}{2} \left[e^{j\frac{2\pi}{4}n} + e^{-j\frac{2\pi}{4}n} \right] x(n)$$

Applying DFT on both sides and using circular frequency shift property.

$$X_2(k) = \frac{1}{2} \left[X((k-1))_4 + X((k+1))_4 \right]$$

Given $X(k) = \{0, (1+j), 1, (1-j)\}$

$$X((k-1))_4 = \{(1-j), 0, (1+j), 1\}$$

$$X((k+1))_4 = \{(1+j), 1, (1-j), 0\}$$

$$\therefore X_2(k) = \frac{1}{2} \begin{bmatrix} 2 & 1 & 2 & 1 \end{bmatrix}$$

$$X_2(k) = \underline{\underline{\{1, 0.5, 1, 0.5\}}}$$

iii) Given $x_3(n) = x((n-1))_4$

taking DFT on both sides and using circular time-shift property.

$$X_3(k) = W_4^k X(k)$$

at $k=0$, $X_3(0) = W_4^0 X(0) = 0$

at $k=1$, $X_3(1) = W_4^1 X(1) = (-j)(1+j) = (1-j)$

at $k=2$, $X_3(2) = W_4^2 X(2) = (-1)(1) = -1$

at $k=3$, $X_3(3) = W_4^3 X(3) = j(1-j) = 1+j$

$$\therefore X_3(k) = \underline{\underline{\{0, 1-j, -1, 1+j\}}}$$

- 35) Let $x(n) = \{1, 2, 3, 4\}$ with $X(k) = \{10, -2+2j, -2, -2-2j\}$. Find the DFT of $x_1(n) = \{1, 0, 2, 0, 3, 0, 4, 0\}$ without actually calculating the DFT.

Soln:- DFT $\{x_1(n)\} = X_1(k) = \sum_{n=0}^{N-1} x_1(n) W_N^{kn}$

Dividing above summation into even and odd parts.

$$X_1(k) = \sum_{n=0}^3 x_1(2n) W_N^{2nk} + \sum_{n=0}^3 x_1(2n+1) W_N^{(2n+1)k}$$

$N=8,$

$$X_1(k) = \sum_{n=0}^3 x_1(2n) W_8^{2nk} + 0 \quad \because \text{odd numbered samples are zero.}$$

$$X_1(k) = \sum_{n=0}^3 x(n) W_{8/2}^{nk}$$

$$X_1(k) = \sum_{n=0}^3 x(n) W_4^{nk} \quad \text{or} \quad X_1(k) = \begin{cases} X(k), & n=0,1,2,3 \\ X(k), & n=4,5,6,7 \end{cases}$$

$$X_1(k) = X(k)$$

$$\text{ie, } X_1(k) = \{10, -2+2j, -2, -2-2j, 10, -2+2j, -2, -2-2j\}$$

Using periodicity property,

$$X_1(4) = X(4) = X(0) \dots\dots$$

- (36) Let $X(k)$ be a 14-point DFT of a length-14 real sequence $x(n)$. The first 8 samples of $X(k)$ are given by

$$X(0) = 12, \quad X(1) = -1 + j3, \quad X(2) = 3 + j4$$

$$X(3) = 1 - j5, \quad X(4) = -2 + j2, \quad X(5) = 6 + j3$$

$$X(6) = -2 - j3, \quad X(7) = 10$$

Determine the remaining samples of $X(k)$.

Evaluate the following functions of $x(n)$ without computing the IDFT of $X(k)$.

(i) $x(0)$ (ii) $x(7)$ (iii) $\sum_{n=0}^{13} x(n)$

(iv) $\sum_{n=0}^{13} e^{j\frac{4\pi}{7}n} x(n)$ (v) $\sum_{n=0}^{13} |x(n)|^2$

Soln:- For a real valued sequence $x(n)$, symmetry property of real valued sequence.

$$X(k) = X^*(N-k) = X^*((-k))_N$$

$$X(8) = X^*(14-8) = X^*(6) = -2 - j3$$

$$X(9) = X^*(14-9) = X^*(5) = 6 - j3$$

$$X(10) = X^*(14-10) = X^*(4) = -2 - j2$$

$$X(11) = X^*(14-11) = X^*(3) = 1 + j5$$

$$X(12) = X^*(14-12) = X^*(2) = 3 - j4$$

$$X(13) = X^*(14-13) = X^*(1) = -1 - j3$$

(i) We know $x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N} km}$; $0 \leq m \leq N-1$

$$N=14, \quad x(m) = \frac{1}{14} \sum_{k=0}^{13} X(k) e^{j\frac{\pi}{7} km}$$

Put $m=0$, $x(0) = \frac{1}{14} \sum_{k=0}^{13} X(k)$

$$= \frac{1}{14} [X(0) + X(1) + X(2) + \dots + X(13)]$$

$$x(0) = \frac{32}{14} = \underline{\underline{2.2857}}$$

(ii) We know, $x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N} km}$

$$N=14, \quad x(m) = \frac{1}{14} \sum_{k=0}^{13} X(k) e^{j\frac{2\pi}{14} km}$$

Put $m=7$, $x(7) = \frac{1}{14} \sum_{k=0}^{13} (-1)^k X(k)$

$$= \frac{-12}{14} = \underline{\underline{-0.8571}}$$

(iii) $X(k) = \sum_{n=0}^{13} x(n) e^{-j\frac{2\pi}{14} kn}$

Put $k=0$, $X(0) = \sum_{n=0}^{13} x(n)$

$$\therefore \sum_{n=0}^{13} x(n) = X(0) = \underline{\underline{12}}$$

iv) The DFT of $e^{j\frac{4\pi}{7}m} x(m)$ or $e^{j\frac{2\pi}{14}4m} x(m)$ is

$$X((k-4))_{14}$$

We know, $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn}$; $0 \leq k \leq N-1$

$$\therefore X((k-4))_{14} = \sum_{n=0}^{13} \left\{ e^{j\frac{4\pi}{7}m} x(m) \right\} e^{-j\frac{2\pi}{14}km}$$

Put $k=0$,

$$X((-4))_{14} = \sum_{n=0}^{13} e^{j\frac{4\pi}{7}m} x(m)$$

$$\begin{aligned} \therefore \sum_{n=0}^{13} e^{j\frac{4\pi}{7}m} x(m) &= X(10) \\ &= \underline{\underline{-2-j2}} \end{aligned}$$

v) From Parseval's Theorem,

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

$$N=14, \quad \sum_{n=0}^{13} |x(n)|^2 = \frac{1}{14} \sum_{k=0}^{13} |X(k)|^2$$

$$= \frac{1}{14} \left[|X(0)|^2 + |X(1)|^2 + \dots \dots \dots + |X(13)|^2 \right]$$

$$= \frac{498}{14} = \underline{\underline{35.5714}}$$